Assessment of Musculoskeletal Function

Initial Interview

- General impression of the patient's health status.
- Gathering subjective data from the patient concerning the onset of the problem and how it has been managed.
- Concurrent health conditions: (eg, diabetes, heart disease, infection, preexisting disability).
- Familial or genetic abnormalities.
- History of medication use and response to analgesics.
- Allergies.
- Use of tobacco, alcohol, and prescription and OTC drugs.
- Learning ability.
- Economic status.
- Current occupation.

Assessment Data

- During the interview and physical assessment, the patient may report:
  - Pain
  - Tenderness
  - Tightness
  - Abnormal sensations

Pain

- **Bone pain** is characteristically described as a dull, deep ache that is "boring" in nature.
- **Muscular pain** is described as soreness or aching and is referred to as muscle cramps.
- **Fracture pain:** sharp and piercing and is relieved by immobilization.
- **Sharp pain** may also result from bone infection with muscle spasm or pressure on a sensory nerve.
- **Pain that increases with activity** may indicate joint sprain or muscle strain.
- **Steadily increasing pain** points to the progression of an infectious process (osteomyelitis), a malignant tumor, or neurovascular complications.
- **Radiating pain** occurs in conditions in which pressure is exerted on a nerve root.

PQRST Method for Pain Assessment

P = Provocation and Palliation	**Q = Quality and Quantity**
What causes it? What makes it better? What makes it worse?	How does it feel, look or sound? How much of it is there?
R = Region and Radiation	**S = Severity and Scale**
Where is it? Does it spread?	Does it interfere with activities? How does it rate on a severity scale of 1 to 10?
T = Timing and Type of Onset	
When did it begin? How often does it occur? Is it sudden or gradual?	

Altered Sensations

- Sensory disturbances are frequently associated with musculoskeletal problems.
- The patient may describe **Paresthesias**, which are burning, tingling sensations or numbness.
- These sensations may be caused by pressure on nerves or by circulatory impairment.
- Soft tissue swelling or direct trauma to these structures can impair their function.

Assessing for Peripheral Nerve Function

- Assessment of peripheral nerve function has two key elements:
  - Evaluation of sensation
  - Evaluation of motion

Peripheral Nerve Function

Nerve	Test of Sensation	Test of Movement
Peroneal nerve	Prick the skin centered between the great and second toe.	Ask the patient to dorsiflex the ankle and extend the toes.

Nerve	Test of Sensation	Test of Movement
Tibial nerve	Prick the medial and lateral surface of the sole.	Ask the patient to plantarflex toes and ankle.

Nerve	Test of Sensation	Test of Movement
Radial nerve	Prick the skin centered between the thumb and	Ask the patient to stretch out the thumb, then the wrist, and then the

	second finger.	fingers at the metacarpal joints.

Nerve	Test of Sensation	Test of Movement
Ulnar nerve	Prick the fat pad at the top of the small finger.	Ask the patient to spread all fingers.

Nerve	Test of Sensation	Test of Movement
Median nerve	Prick the top or distal surface of the index finger.	Ask the patient to touch the thumb to the little finger. Also observe whether the patient can flex the wrist.

Posture

- The normal curvature of the spine is convex through the thoracic portion and concave through the cervical and lumbar portions.

Common deformities of the spine include:

- **Kyphosis**, an increased forward curvature of the thoracic spine.
- **Lordosis**, or swayback, an exaggerated curvature of the lumbar spine.
- **Scoliosis**, a lateral curving deviation of the spine.
- **Kyphosis** is frequently seen in elderly patients with osteoporosis and in some patients with neuromuscular diseases.
- **Scoliosis** may be congenital, or the result of damage to the paraspinal muscles, as in polio.
- **Lordosis** is frequently seen during pregnancy as the woman adjusts her posture in response to changes in her center of gravity.

Neurovascular Status

- It is important for the nurse to perform frequent neurovascular assessments because of the risk for tissue and nerve damage.
- Complication that the nurse needs to be alert for is compartment syndrome.
- This major neurovascular problem is caused by pressure within a muscle compartment that increases to such an extent that microcirculation diminishes, leading to nerve and muscle anoxia and necrosis.
- Function can be permanently lost if the anoxic situation continues for longer than 6 hours.
- Assessment of neurovascular status is CMS (circulation, motion, and sensation).

Indicators of Peripheral Neurovascular Dysfunction

- **Circulation**
  - Color: Pale, cyanotic, or mottled
  - Temperature: Cool
  - Capillary refill: More than 3 seconds

- **Motion**
  - Weakness
  - Paralysis
- **Sensation**
  - Paresthesia
  - Unrelenting pain
  - Pain on passive stretch
  - Absence of feeling

Imaging Procedures

X-ray Studies

- Bone x-rays determine bone density, texture, erosion, and changes in bone relationships.
- Joint x-rays reveal fluid, irregularity, spur formation, narrowing, and changes in the joint structure.
- Serial x-rays may be indicated to determine if healing of a fractured bone is progressing normally or to determine if a bone affected by a degenerative disease (eg, osteoarthritis) is responding to prescribed therapy.
- After being positioned for the study, the patient must remain still while the x-rays are taken.

Computed Tomography

- A computed tomography (CT) scan shows in detail of a bone and can reveal tumors of the soft tissue or injuries to the ligaments or tendons.
- It is used to identify the location and extent of fractures in areas that are difficult to evaluate (eg, acetabulum).
- CT studies, which may be performed with or without the use of contrast agents, last about 1 hour.
- The patient must remain still during the procedure.

Magnetic Resonance Imaging

- Magnetic resonance imaging (MRI) is a noninvasive imaging technique that uses magnetic fields to demonstrate abnormalities (ie, tumors or narrowing of tissue pathways through bone) of soft tissues such as muscle, tendon, cartilage, nerve, and fat.
- Because an electromagnet is used, patients with any metal implants, clips, or pacemakers are not candidates for MRI.

- To enhance visualization of anatomic structures, intravenous (IV) contrast agent may be used.
- Patients who experience claustrophobia may be unable to tolerate the closed MRI equipment without sedation.
- Jewelry, hair clips, hearing aids, credit cards with magnetic strips, and other metal-containing objects must be removed before the MRI is performed.

Arthrography

- Arthrography is useful in identifying acute or chronic tears of the joint capsule or supporting ligaments of the knee, shoulder, ankle, hip, or wrist.
- A radiopaque contrast agent or air is injected into the joint cavity to outline soft tissue structures and the contour of the joint.
- The joint is put through its range of motion to distribute the contrast agent while a series of x-rays is obtained.
- If a tear is present, the contrast agent leaks out of the joint and is evident on the x-ray image.
- After an arthrogram, a compression elastic bandage is applied as prescribed and the joint is usually rested for 12 hours.
- The nurse provides additional comfort measures (mild analgesia, ice) as appropriate.
- The nurse explains to the patient that it is normal to experience clicking or crackling in the joint for a day or two after the procedure.

Bone Densitometry

- Bone densitometry is used to estimate bone mineral density (BMD).
- This can be performed through the use of x-rays or ultrasound.

- Dual-energy x-ray absorptiometry (DEXA) determines bone mineral density at the wrist, hip, or spine to estimate the extent of osteoporosis and to monitor a patient's response to treatment for osteoporosis.
- Heel bone sonometry (ultrasound) and peripheral DEXAs are effective tools for osteoporosis.
- Hip BMD testing is considered the most accurate test for osteoporosis and for predicting risk of hip fracture.
- Evidence-based guidelines espouse using hip BMD as the first-line test for osteoporosis.

Nuclear Studies

Bone Scan

- A bone scan is performed to detect metastatic and primary bone tumors, osteomyelitis, certain fractures, and aseptic necrosis.
- A bone-seeking radioisotope is injected IV.
- The scan is performed 2 to 3 hours after the injection.
- At this point, distribution and concentration of the isotope in the bone are measured.
- The degree of nuclide uptake is related to the metabolism of the bone.
- An increased uptake of isotope is seen in primary skeletal disease (osteosarcoma), metastatic bone disease, inflammatory skeletal disease(osteomyelitis), and certain types of fractures.

Scan A shows hot spots (dark areas) in both knees, a sign of arthritis, and a possible fracture in the second toe of the right foot. Otherwise it shows normal bone metabolism. Scan B shows numerous bone hot spots, a result of cancer that has spread to multiple locations.

Endoscopic Studies: Arthroscopy

- Arthroscopy is a procedure that allows direct visualization of a joint to diagnose joint disorders.
- Treatment of tears, defects, and disease processes may be performed through the arthroscope.
- The procedure is performed in the operating room under sterile conditions; injection of a local anesthetic into the joint or general anesthesia is used.
- A large-bore needle is inserted, and the joint is distended with saline.
- The arthroscope is introduced, and joint structures, synovium, and articular surfaces are visualized.
- After the procedure, the puncture wound is closed with adhesive strips or sutures and covered with a sterile dressing.
- Complications are rare but may include infection, hemarthrosis, neurovascular compromise, thrombophlebitis, stiffness, effusion, adhesions, and delayed wound healing.
- **Nursing Interventions:**
  - The joint is wrapped with a compression dressing to control swelling.
  - Ice may be applied to control edema and discomfort.
  - The joint is kept extended and elevated to reduce swelling.
  - Monitor neurovascular status.
  - Administer prescribed analgesics to control discomfort.

- Explains to the patient and family the symptoms (eg, swelling, numbness, cool skin) to monitor in order to determine whether complications are occurring.

Arthrocentesis

- Arthrocentesis (joint aspiration) is carried out to obtain synovial fluid for purposes of examination or to relieve pain due to effusion.
- Examination of synovial fluid is helpful in the diagnosis of septic arthritis and other inflammatory arthropathies and reveals the presence of hemarthrosis (bleeding into the joint cavity).
- Normally, synovial fluid is clear, pale, straw-colored, and scanty in volume.
- Using aseptic technique, the physician inserts a needle into the joint and aspirates fluid.
- Anti-inflammatory medications may be injected into the joint.
- A sterile dressing is applied after aspiration.
- There is a risk of infection after this procedure.

Musculoskeletal Care Modalities

The Patient in a Cast

- It is a rigid device applied to immobilize the injured bones and promote healing.
- It is applied to immobilize the joint above and below the fractured bone so that the bone will not move during healing.
- These are applied on clients who have relatively stable fractures.

Casting materials

- The outside, or hard part of the cast, is made from two different kinds of casting materials.
  - Plaster - white in color.
  - Fiberglass - comes in a variety of colors, patterns, and designs.
- Cotton and other synthetic materials are used to line the inside of the cast to make it soft and to provide padding around bony areas, such as the wrist or elbow.

Types of casts

Type of Cast	Location	Uses
Short arm cast:	Applied below the elbow to the hand.	Forearm or wrist fractures. Also used to hold the forearm or wrist muscles and tendons in place after surgery.
Long arm cast:	Applied from the upper arm to the hand.	Upper arm, elbow, or forearm fractures. Also used to hold the arm or elbow muscles and tendons in place after surgery.
Arm cylinder cast:	Applied from the upper arm to the wrist.	To hold the elbow muscles and tendons in place after a dislocation or surgery.

Type of Cast	Location	Uses
Shoulder spica cast:	Applied around the trunk of the body to the shoulder, arm, and hand.	Shoulder dislocations or after surgery on the shoulder area.
Minerva cast:	Applied around the neck and trunk of the body.	After surgery on the neck or upper back area.
Short leg cast:	Applied to the area below the knee to the foot.	Lower leg fractures, severe ankle sprains/strains, or fractures. Also used to hold the leg or foot muscles and tendons in place after surgery to allow healing.

Leg cylinder cast:	Applied from the upper thigh to the ankle.	Knee, or lower leg fractures, knee dislocations, or after surgery on the leg or knee area.
Long leg cast:	Applied from the upper thigh to the foot.	Stable fractures of the tibia and fibula, fractures of the patella, stable fractures of the distal shaft of the femur

Type of Cast	Location	Uses
Unilateral hip spica cast:	Applied from the chest to the foot on one leg.	Thigh fractures. Also used to hold the hip or thigh muscles and tendons in place after surgery to allow healing.
One and one-half hip spica cast:	Applied from the chest to the foot on one leg to the knee of the other leg. A bar is placed between both legs to keep the hips and legs immobilized.	Thigh fracture. Also used to hold the hip or thigh muscles and tendons in place after surgery to allow healing.
Bilateral long leg hip spica cast:	Applied from the chest to the feet. A bar is placed between both legs to keep the hips and legs immobilized.	Pelvis, hip, or thigh fractures. Also used to hold the hip or thigh muscles and tendons in place after surgery to allow healing.

Type of Cast	Location	Uses
Short leg hip spica cast:	Applied from the chest to the thighs or knees.	To hold the hip muscles and tendons in place after surgery to allow healing.

Type of Cast	Location	Uses
Abduction boot cast:	Applied from the upper thighs to the feet. A bar is placed between both legs to keep the hips and legs immobilized.	To hold the hip muscles and tendons in place after surgery to allow healing.

Potential Complications

Compartment Syndrome

- Compartment syndrome occurs when there is increased tissue pressure within a limited space (e.g., cast, muscle compartment) that compromises the circulation and the function of the tissue confined area. within the

Management

- To relieve the pressure, the cast must be bivalved (cut in half longitudinally) while maintaining alignment
- The extremity must be elevated no higher than heart level to ensure arterial perfusion.
- If pressure is not relieved and circulation is not restored, a fasciotomy may be necessary to relieve the pressure within the muscle compartment.

Pressure Ulcers

- Pressure of the cast on soft tissues may cause tissue anoxia and pressure ulcers.
- Lower extremity sites most susceptible to pressure ulcers are the heel, malleoli, dorsum of the foot, head of the fibula, and anterior surface of the patella.
- Usually, the patient with a pressure ulcer reports pain and tightness in the area.
- A warm area on the cast suggests underlying tissue erythema.
- The drainage may stain the cast and emit an odor.
- To inspect the pressure ulcer area, the physician may bivalve the cast or cut an opening (window) in the cast.
- The portion of the cast is replaced and held in place by an elastic compression dressing or tape.

Disuse Syndrome

- While in a cast, the patient needs to learn to tense or contract muscles (isometric muscle contraction) without moving the part.
- This helps reduce muscle atrophy and maintain muscle strength.
- The nurse teaches the patient with a leg cast to “push down” the knee and teaches the patient in an arm cast to “make a fist.”
- Isometric exercises should be performed hourly while the patient is awake.

Care of the Patient with an Arm Cast

- To control swelling, the immobilized arm is elevated.
- When the patient is lying down, the arm is elevated so that each joint is positioned higher than the preceding proximal joint.
- A sling may be used when the patient ambulates.
- To prevent pressure on the cervical spinal nerves, the sling should distribute the supported weight over a large area and not on the back of the neck.
- The nurse encourages the patient to remove the arm from the sling and elevate it frequently.
- Circulatory disturbances in the hand may become apparent with signs of cyanosis, swelling, and an inability to move the fingers.
- One serious effect of impaired circulation in the arm is Volkmann's contracture, a specific type of compartment syndrome. Contracture of the fingers and wrist occurs as the result of obstructed arterial blood flow to the forearm and hand.
- Permanent damage develops within a few hours if action is not taken

Care of the Patient with a Leg Cast

- The patient's leg must be supported on pillows to heart level to control swelling, and ice packs should be applied as prescribed over the fracture site for 1 or 2 days.
- The patient is taught to elevate the casted leg when seated.
- The patient should also assume a recumbent position several times a day with the casted leg elevated to promote venous return and control swelling.
- The nurse assesses circulation by observing the color, temperature, and capillary refill of the exposed toes.
- Nerve function is assessed by observing the patient's ability to move the toes and by asking about the sensations in the foot.

- Numbness, tingling, and burning may be caused by peroneal nerve injury from pressure at the head of the fibula.

Care of the Patient with a Body or Spica Cast

- Body casts are used to immobilize the spine.
- Hip spica casts are used for some femoral fractures and after some hip joint surgeries, and shoulder spica casts are used for some humeral neck fractures.
- Nursing responsibilities include preparing and positioning the patient, assisting with skin care and hygiene.
- Explaining the procedure helps reduce the patient's apprehension about being encased in a large cast.
- The nurse reassures the patient that several people will provide care during the application that support for the injured area will be adequate, and that care providers will be as gentle as possible.
- The nurse turns the patient to a prone position, twice daily if tolerated, to provide postural drainage of the bronchial tree and to relieve pressure on the back.
- A small pillow under the abdomen enhances comfort.
- The nurse inspects the skin around the edges of the cast frequently for signs of irritation.
- The nurse can inspect some of the skin under the cast by pulling the skin taut and using a flashlight.
- The skin can be bathed and massaged by reaching under the cast edges with the fingers.

Cast syndrome

Patients immobilized in large casts may develop cast syndrome that may include psychological or physiological manifestations.

- **Psychological manifestations**
  - The psychological component is similar to a claustrophobic reaction.
  - **S&S**
    - The patient exhibits an acute anxiety reaction characterized by behavioral changes and autonomic responses (eg, increased respiratory rate, diaphoresis, dilated pupils, increased heart rate, and elevated blood pressure).
  - **Management**
    - The nurse needs to recognize the anxiety reaction and provide an environment in which the patient feels secure.
- **Physiological manifestations**
  - The physiological cast syndrome responses (superior mesenteric artery syndrome) are associated with immobility in a body cast.
  - With decreased physical activity, gastrointestinal motility decreases, intestinal gases accumulate, intestinal pressure increases, and ileus may occur.
  - **S&S**
    - The patient exhibits abdominal distention, abdominal discomfort, nausea, and vomiting.
  - **Management**
    - As with other instances of adynamic ileus, the patient is treated conservatively with decompression (nasogastric intubation connected to suction) and intravenous (IV) fluid therapy until gastrointestinal motility is restored.

- After the ileus resolves and bowel sounds resume, the patient gradually resumes an oral diet.
- The nurse monitors the patient in a large body cast for potential cast syndrome, noting bowel sounds every 4 to 8 hours, and reports distention, nausea, and vomiting to the physician.

The Patient with an External Fixator

- External fixators are used to manage open fractures with soft tissue damage.
- They provide stable support for severe comminuted fractures while permitting active treatment of damaged soft tissues.
- Complicated fractures of the humerus, forearm, femur, tibia, and pelvis are managed with external skeletal fixators.
- The fracture is reduced, aligned, and immobilized by a series of pins inserted in the bone.
- Pin position is maintained through attachment to a portable frame.
- The fixator facilitates patient comfort, early mobility, and active exercise so complications due to disuse and immobility are minimized.
- The fixator is removed after the soft tissue heals.
- The fracture may require additional stabilization by a cast or molded orthosis while healing.

Nursing Interventions

- It is important to prepare the patient psychologically for application of the external fixator.
- Reassurance that the discomfort associated with the device is minimal.
- After the external fixator is applied, the extremity is elevated to reduce swelling.
- If there are sharp points on the fixator or pins, they are covered with tape to prevent device-induced injuries.

- The nurse monitors the neurovascular status of the extremity every 2 to 4 hours.
- The nurse must be alert for potential problems caused by pressure from the device on the skin, nerves, or blood vessels and for the development of compartment syndrome.
- Some serous drainage from the pin sites is to be expected.
- The nurse carries out pin care as prescribed to prevent pin tract infection.
- This typically includes cleaning each pin site separately one or two times a day with cotton-tipped applicators soaked in chlorhexidine solution.
- If signs of infection are present or if the pins or clamps seem loose, the nurse notifies the physician.
- The nurse encourages isometric and active exercises as tolerated.

The Ilizarov

- The Ilizarov external fixator is a special device used to correct angulation and rotational defects, to treat nonunion (failure of bone fragments to heal), and to lengthen limbs.

The Patient in Traction

- Traction is the application of a pulling force to a part of the body.
- Traction is used to minimize muscle spasms; to reduce, align, and immobilize fractures; to reduce deformity; and to increase space between opposing surfaces.
- Traction must be applied in the correct direction and magnitude to obtain its therapeutic effects.
- The effects of traction are evaluated with x-ray studies, and adjustments are made if necessary.
- Traction is used primarily as a short-term intervention until other modalities, such as external or internal fixation, are possible.
- This reduces the risk of disuse syndrome and minimizes the length of hospitalization.

Principles of Effective Traction

- Traction must be continuous to be effective in reducing and immobilizing fractures.
- Skeletal traction is never interrupted.
- Weights are not removed unless intermittent traction is prescribed.
- The patient must be in good body alignment in the center of the bed when traction is applied.
- Ropes must be unobstructed.
- Weights must hang freely and not rest on the bed or floor.
- Knots in the rope or the footplate must not touch the pulley or the foot of the bed.

Types of traction

Skin Traction

- Skin traction is accomplished by weights that pull on tape, sponge rubber or plastic materials attached to the skin.
- TRACTION on the SKIN, TRANSMITS traction to the musculoskeletal structures.
- Skin traction is used to control muscle spasms and to immobilize an area before surgery.
- The amount of weight applied must not exceed the tolerance of the skin.
- No more than 2 to 3.5 kg of traction can be used on an extremity.
- Pelvic traction is usually 4.5 to 9 kg, depending on the weight of the patient.

Monitoring and Managing Potential Complications

Skin Breakdown

- During the initial assessment, the nurse identifies fragile skin.
- The nurse also monitors the status of the skin in contact with tape.
- The nurse performs the following procedures to monitor and prevent skin breakdown:
- Remove the foam boots to inspect the skin, the ankle, and the achilles tendon three times a day.
- A second nurse is needed to support the extremity during the inspection and skin care.
- Palpate the area of the traction tapes daily to detect underlying tenderness.
- Provide back care at least every 2 hours to prevent pressure ulcers.
- Use air mattress.

Nerve Damage

- When traction is applied to the lower extremity, care must be taken to avoid pressure on the peroneal nerve at the point at which it passes around the neck of the fibula just below the knee.
- Pressure at this point can cause footdrop.
- The nurse questions the patient about sensation and asks the patient to move the toes and foot.
- Dorsiflexion of the foot demonstrates function of the peroneal nerve. Weakness of dorsiflexion or foot movement and inversion of the foot might indicate pressure on the common peroneal nerve.
- The following are important points to keep in mind when caring for the patient in traction:
  - Regularly assess sensation and motion.
  - Immediately investigate any complaint of a burning sensation under the traction bandage or boot.
  - Promptly report altered sensation or impaired motor function.

Circulatory Impairment

- After skin traction is applied, the nurse assesses circulation of the foot or hand within 15 to 30 minutes and then every 1 to 2 hours.
- Circulatory assessment consists of the following:
  - Peripheral pulses, color, capillary refill, and temperature of the fingers or toes
  - Indicators of DVT, including unilateral calf tenderness, warmth, redness, and swelling

Skeletal Traction

- Skeletal traction is applied directly to the bone.
- This method of traction is used occasionally to treat fractures of the femur, the tibia, and the cervical spine.
- The orthopedic surgeon applies skeletal traction, using surgical asepsis.

- The insertion site is prepared with a surgical scrub agent such as povidone–iodine solution.
- A local anesthetic is administered at the insertion site and periosteum.
- The surgeon makes a small skin incision and drills the sterile pin or wire through the bone.
- The patient feels pressure during this procedure and possibly some pain when the periosteum is penetrated.
- The weights are attached to the pin or wire bow by a rope-and-pulley system that exerts the appropriate amount and direction of pull for effective traction.
- Skeletal traction frequently uses 7 to 12 kg to achieve the therapeutic effect.
- Often, skeletal traction is balanced traction, which supports the affected extremity, allows for some patient movement, and facilitates patient independence and nursing care while maintaining effective traction
- When skeletal traction is discontinued, the extremity is gently supported while the weights are removed.
- Internal fixation, casts, or splints are then used to immobilize and support the healing bone.

Nursing Interventions

- **Maintaining Effective Traction**
- When skeletal traction is used, the nurse checks the traction apparatus to see that the ropes are in the wheel grooves of the pulleys that the weights hang freely, and that the knots in the rope are tied securely.
- The nurse also evaluates the patient's position, because slipping down in bed results in ineffective traction.
- Maintaining Positioning
- Preventing Skin Breakdown
- Monitoring Neurovascular Status

- Providing Pin Site Care
- Promoting Exercise

Nursing Alert

- The nurse must never remove weights from skeletal traction unless a life-threatening situation occurs. Removal of the weights completely defeats their purpose and may result in injury to the patient

Joint Replacement

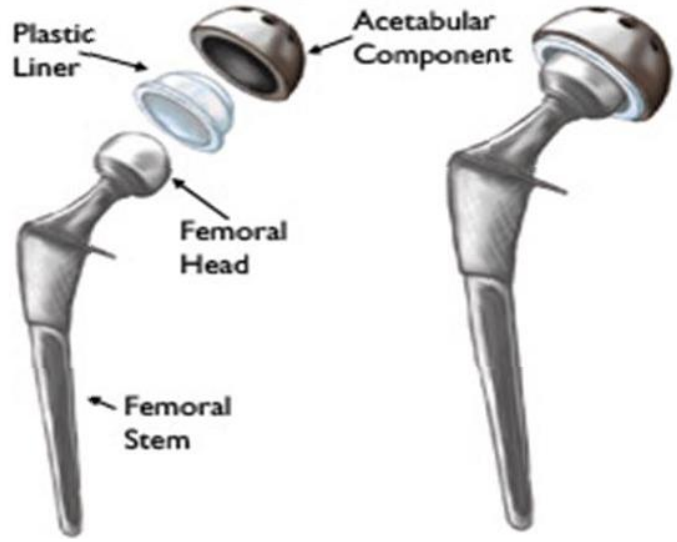
- Patients with severe joint pain and disability may undergo joint replacement.
- Joints frequently replaced include the hip, knee.
- Most joint replacements consist of metal and high-density polyethylene components.
- With joint replacement, excellent pain relief is obtained in most patients.
- Return of motion and function depends on preoperative soft tissue condition, soft tissue reactions, and general muscle strength.
- Early failure of joint replacement is associated with excessive activity and preoperative joint and bone pathology.

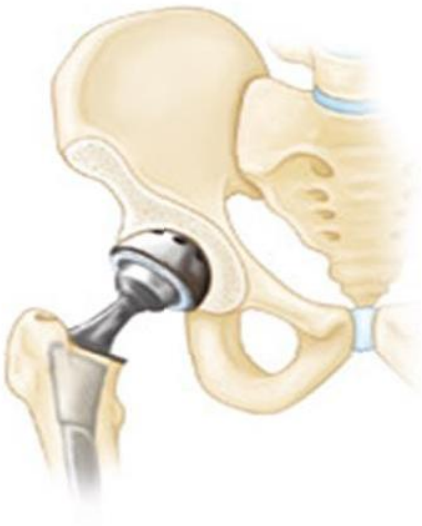
Nursing Interventions

- Assessment of the patient.
- Preoperatively, it is important to assess the neurovascular status of the extremity undergoing joint replacement.
- Postoperative assessment data are compared with preoperative assessment data to identify changes and deficits. For example, an absent pulse postoperatively is of concern unless the pulse was also absent preoperatively.
- Nerve palsy could occur as a result of surgery.

Total Hip Replacement

- A total hip replacement is a surgical procedure whereby the diseased cartilage and bone of the hip joint is surgically replaced with artificial materials. The normal hip joint is a ball and socket joint.
- Total hip replacement is the replacement of a severely damaged hip with an artificial joint (prosthesis).
- Indications for this surgery include arthritis, femoral neck fractures, and conditions resulting from congenital hip disease.





Complications

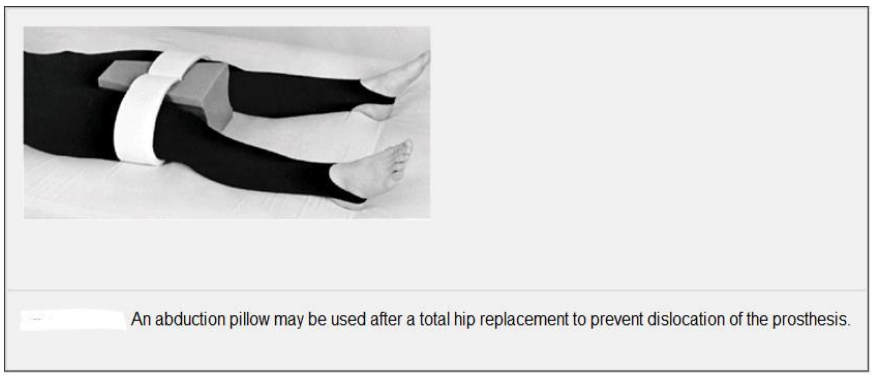
- Dislocation of the hip prosthesis
- Excessive wound drainage
- Thromboembolism
- Infection
- Heel pressure ulcer
- Heterotopic ossification (formation of bone in the periprosthetic space)
- Avascular necrosis (bone death caused by loss of blood supply)
- Loosening of the prosthesis.

Preventing Dislocation of the Hip Prosthesis

- Maintenance of the femoral head component in the acetabular cup is essential.
- The nurse teaches the patient about positioning the leg in abduction, which helps prevent dislocation of the prosthesis.
- The use of an abduction splint, a wedge pillow
- When the nurse turns the patient in bed, it is important to keep the operative hip in abduction.
- The patient's hip is never flexed more than 90 degrees.
- To prevent hip flexion, the nurse does not elevate the head of the bed more than 60 degrees.
- The nurse encourages the patient to keep the affected hip in extension.
- When sitting, the patient's hips should be higher than the knees.
- The patient's affected leg should not be elevated when sitting.
- The patient may flex the knee.

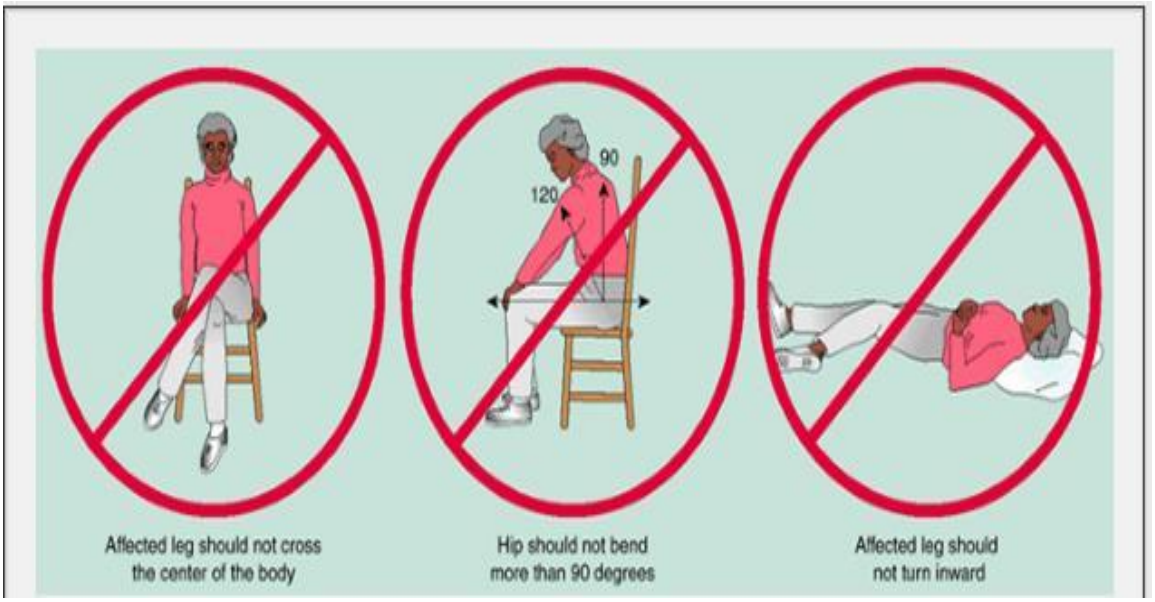
Patient Education

- **Methods for avoiding displacement include the following:**
- Keep the knees apart at all times.
- Put a pillow between the legs when sleeping.
- Never cross the legs when seated.
- Avoid bending forward when seated in a chair.
- Avoid bending forward to pick up an object on the floor.
- Use a high-seated chair and a raised toilet seat.
- Do not flex the hip to put on clothing such as pants, stockings, socks, or shoes.



An abduction pillow may be used after a total hip replacement to prevent dislocation of the prosthesis.

- **Positions to avoid after total hip replacement**

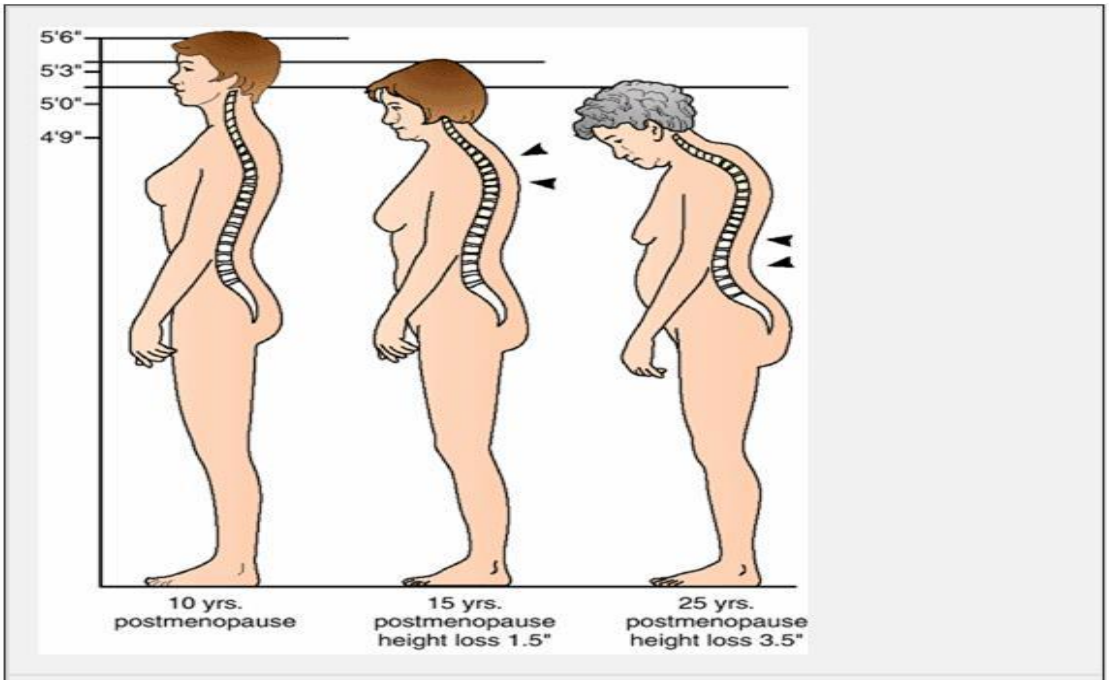


Management of Patients with Musculoskeletal Disorders

Osteoporosis

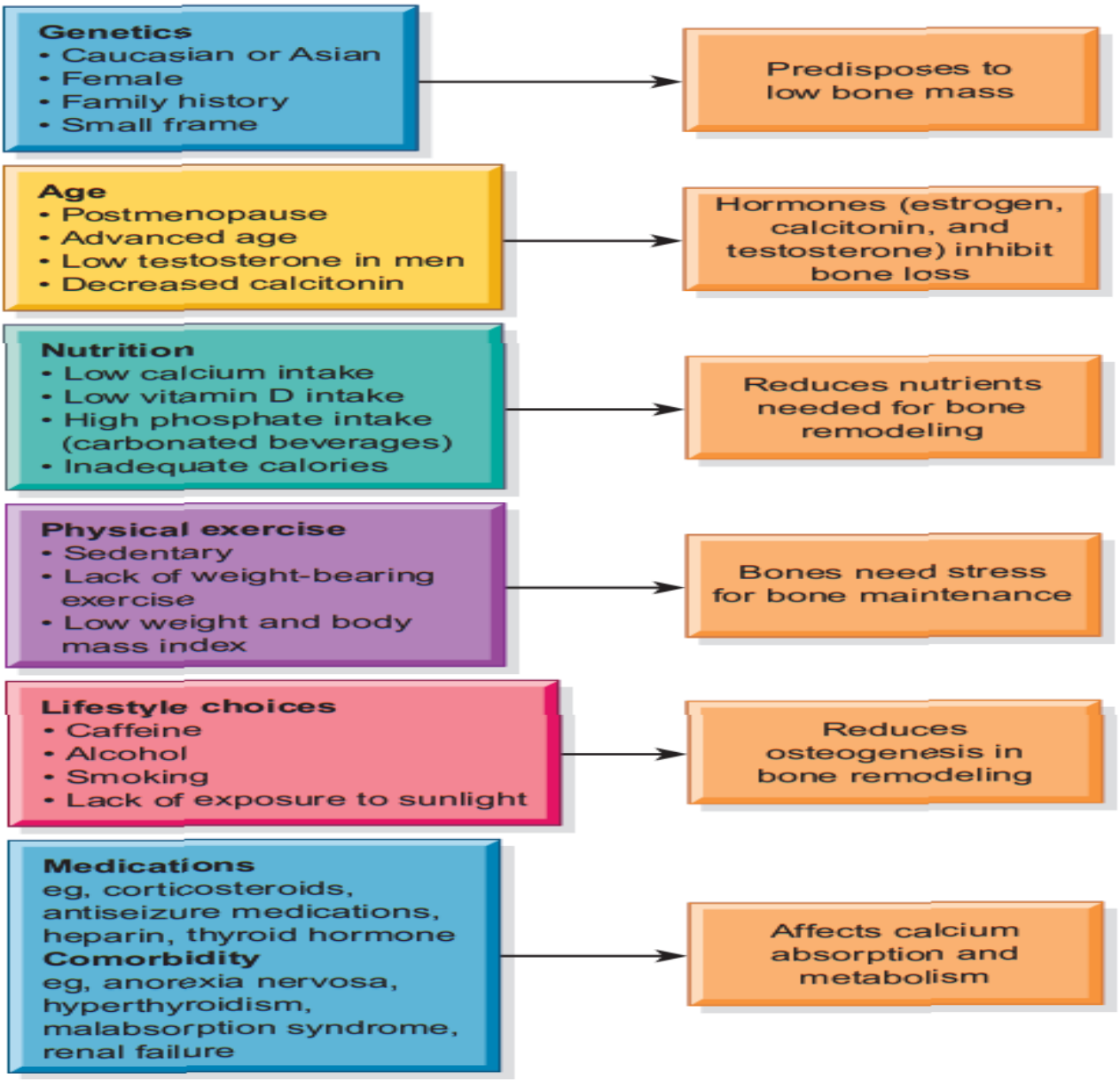
- Characteristics of osteoporosis include a reduction of bone density and a change in bone structure.
- The normal homeostatic bone turnover is altered
- The rate of bone resorption is greater than the rate of bone formation, resulting in a reduced total bone mass.
- Suboptimal bone mass development in children and teens contributes to the development of osteoporosis.
- With osteoporosis, the bones become progressively porous, brittle, and fragile; they fracture easily under stresses that would not break normal bone.
- Osteoporosis frequently results in compression fractures of the thoracic and lumbar spine, fractures of the neck of the femur.
- These fractures may be the first clinical manifestation of osteoporosis.
- Fracture risk increases with increasing age.
- Multiple compression fractures of the vertebrae result in skeletal deformity (kyphosis).

- Frequently, postmenopausal women lose height from vertebral collapse.
- The postural changes result in relaxation of the abdominal muscles and a protruding abdomen.
- The deformity may also produce pulmonary insufficiency.
- Many patients complain of fatigue.



Loss of height associated with osteoporosis and aging

Risk factors for osteoporosis, and the effects of these factors on bone



Prevention

- Up to 90% of peak adult bone mass is achieved between the ages of 18 and 20 years.
- Primary osteoporosis occurs in women after menopause (usually between the ages of 45 and 55 years)
- Early identification of at-risk teenagers and young adults, increased calcium intake, participation in regular weight-bearing exercise, and modification of lifestyle (reduced use of caffeine, cigarettes, carbonated soft drinks, and alcohol).
- Secondary osteoporosis is the result of medications or other conditions and diseases that affect bone metabolism.
- Specific disease states (celiac disease) and medications (corticosteroids) that place patients at risk need to be identified and therapies instituted to reverse the development of osteoporosis.

Assessment and Diagnostic Findings

- Routine x-rays
- Osteoporosis is diagnosed by Bone Densitometry: i.e. Dual-energy x-ray absorptiometry (DEXA).
- DEXA of the wrist, hip, or spine are used to screen for osteoporosis.
- Laboratory studies (serum calcium, serum phosphate, serum alkaline phosphatase, urine calcium excretion, hematocrit, erythrocyte sedimentation rate [ESR]).

Management

- A diet rich in calcium and vitamin D throughout life.
- Such a diet includes three glasses of skim or whole vitamin D–enriched milk or other foods high in calcium.
- calcium supplement with Vitamin D may be prescribed and taken with meals or with a beverage high in vitamin C to promote absorption.
- Common side effects of calcium supplements are abdominal distention and constipation.
- From 20 to 30 minutes of aerobic exercise, 3 days or more a week, is recommended.
- Smoking cessation
- Alcohol and caffeine consumption in moderation
- Regular weight-bearing exercise regimen

Pharmacologic Therapy

- Estrogen replacement decreases bone resorption and increases bone mass.
- Studies have demonstrated greater risks including: strokes, venous thromboemboli, and breast cancer, than the benefits.
- Selective estrogen receptor modulators.
- Bisphosphonates reduce spine and hip fractures associated with osteoporosis through increasing bone mass and decreasing bone loss by inhibiting osteoclast function.
- Side effects of bisphosphonates include: gastrointestinal symptoms (dyspepsia, nausea, flatulence, diarrhea, constipation, esophageal ulcers, gastric ulcers.
- Calcitonin directly inhibits osteoclasts, thereby reducing bone loss.
- Calcitonin is administered by nasal spray or by subcutaneous or intramuscular injection.

- Side effects include: nasal irritation, flushing, gastrointestinal disturbances, and urinary frequency.
- In addition, antilipid medications, such as statins, reduce the incidence of fractures.
- The statins promote bone growth, thereby preventing the development of osteoporosis.

Musculoskeletal Infections

Osteomyelitis

- Osteomyelitis is an infection of the bone.
- The bone becomes infected in one of three ways:
  - Extension of soft tissue infection (infected pressure or vascular ulcer, incisional infection)
  - Direct bone contamination from bone surgery, open fracture, or traumatic injury (eg, gunshot wound)
  - Hematogenous (bloodborne) spread from other sites of infection (infected tonsils, infected teeth, upper respiratory infections).
- Bone infections are more difficult to eradicate than soft tissue infections because the infected bone is mostly avascular and not accessible to the body's natural immune response and decreased penetration by antibiotics.

Risk factors

- Poorly nourished
- Elderly
- Obese
- Impaired immune systems
- Chronic illnesses (diabetes, rheumatoid arthritis)
- Long-term corticosteroid therapy.
- Postoperative surgical wound infections occur within 30 days after surgery.

Pathophysiology

- Between 70% and 80% of bone infections are caused by Staphylococcus aureus.
- The initial response to infection is inflammation, increased vascularity, and edema.

- After 2 or 3 days, thrombosis of the blood vessels occurs in the area, resulting in ischemia with bone necrosis.
- The infection may spread into adjacent soft tissues and joints.
- Unless the infective process is treated promptly, a bone abscess forms.
- The resulting abscess cavity contains dead bone tissue (the sequestrum).
- New bone growth (the involucrum) forms and surrounds the sequestrum.
- Although healing appears to take place, a chronically infected sequestrum remains and produces recurring abscesses throughout the patient's life.
- This is referred to as chronic osteomyelitis.

Clinical Manifestations

- When the infection is bloodborne (chills, high fever, rapid pulse, general malaise).
- The infected area becomes painful, swollen, and extremely tender.
- The patient may describe a constant, pulsating pain that intensifies with movement as a result of the pressure of the collecting pus.
- The area is swollen, warm, painful, and tender to touch.
- The patient with chronic osteomyelitis presents with a continuously draining sinus or experiences recurrent periods of pain, inflammation, swelling, and drainage.

Assessment and Diagnostic Findings

- In acute osteomyelitis, early x-ray findings demonstrate soft tissue swelling.
- In about 2 weeks, areas of irregular decalcification, bone necrosis, periosteal elevation, and new bone formation are evident.
- Magnetic resonance imaging
- Blood studies reveal leukocytosis and an elevated ESR.
- Wound and blood culture studies are performed to identify appropriate antibiotic therapy.

- Anemia, associated with chronic infection, may be evident.
- The abscess is cultured to determine the infective organism and antibiotic.

Prevention

- Elective orthopedic surgery should be postponed if the patient has a current infection (eg, urinary tract infection, sore throat)
- During orthopedic surgery, careful attention is paid to the surgical environment and techniques.
- Prophylactic antibiotics.
- Urinary catheters and drains are removed as soon as possible to decrease the incidence of hematogenous spread of infection.
- Aseptic postoperative wound care.
- Prompt management of soft tissue infections reduces extension of infection to the bone.
- When patients who had a joint replacement surgery are going to have invasive procedures (eg, cystoscopy), prophylactic antibiotics are frequently recommended.

Management

- Antibiotic therapy depends on the results of blood and wound cultures.
- IV antibiotic therapy penicillin or cephalosporin for 3 to 6 weeks.
- After the infection controlled, the antibiotic administered orally for up to 3 months.
- General supportive measures (eg, hydration, diet high in vitamins and protein, correction of anemia).
- The area affected with osteomyelitis is immobilized to decrease discomfort and to prevent pathologic fracture.
- Warm wet soaks for 20 minutes several times a day may increase circulation.

Surgical Management

- Surgical débridement
- The infected bone is surgically exposed, the purulent and necrotic material is removed, and the area is irrigated with sterile saline solution.
- Wound irrigation using sterile saline solution may be performed for 7 to 8 days.
- Direct application of antibiotics for 2 to 4 weeks.
- A sequestrectomy (removal of enough involucrum to enable the surgeon to remove the sequestrum)

Management of Patients with Musculoskeletal Trauma

Contusions

- A contusion is a soft tissue injury produced by blunt force, such as a blow, kick, or fall.
- Many small blood vessels rupture and bleed into soft tissues (ecchymosis, or bruising).
- A hematoma develops when the bleeding is sufficient to cause collection of blood.
- Local symptoms (pain, swelling, and discoloration) are controlled with intermittent application of cold packs.
- Most contusions resolve in 1 to 2 weeks.

Strain

- A strain, or a "pulled muscle," is an injury to a musculotendinous unit caused by overuse, overstretching, or excessive stress.

Sprain

- A sprain is an injury to the ligaments and supporting muscle fibers that surround a joint.
- Blood vessels rupture and edema occurs, the joint is tender, and movement of the joint becomes painful.
- The degree of disability and pain increases during the first 2 to 3 hours after the injury because of the associated swelling and bleeding.

Management

RICE

- Rest Ice Compression Elevation
- Rest prevents additional injury and promotes healing.

- Intermittent application of moist or dry cold packs for 20 to 30 minutes during the first 24 to 48 hours after injury produces vasoconstriction, which decreases bleeding, edema, and discomfort.
- An elastic compression bandage controls bleeding, reduces edema, and provides support for the injured tissues.
- Elevation controls the swelling.
- If the sprain or strain is third-degree, surgical repair or immobilization by cast may be necessary so that the joint will not lose its stability.
- After the acute inflammatory stage (24 to 48 hours after injury), heat may be applied intermittently (for 15 to 30 minutes, four times a day) to relieve muscle spasm and to promote vasodilation, absorption, and repair.
- Depending on the severity of injury, progressive passive and active exercises may begin in 2 to 5 days.
- Severe sprains and strains may require 1 to 3 weeks of immobilization before exercises are initiated.
- Strains and sprains take weeks or months to heal, because ligaments and tendons are relatively avascular.
- Splinting may be used to prevent reinjury.

Joint Dislocations

- A dislocation of a joint is a condition in which the articular surfaces of the bones forming the joint are no longer in anatomic contact.
- A subluxation is a partial dislocation of the articulating surfaces.
- The bones are literally “out of joint.”
- Traumatic dislocations are orthopedic emergencies because the associated joint structures, blood supply, and nerves are distorted and severely stressed.
- If the dislocation is not treated promptly, avascular necrosis and nerve palsy may occur.

- Dislocations may be congenital, or present at birth, spontaneous or pathologic, caused by disease, or traumatic, resulting from injury in which the joint is disrupted by force.

Signs and Symptoms

- Pain
- Change in contour of the joint
- Change in the length of the extremity
- Loss of normal mobility
- Change in the axis of the dislocated bones.
- X-rays confirm the diagnosis and reveal any associated fracture.

Management

- The affected joint needs to be immobilized while the patient is transported to the hospital.
- The dislocation is promptly reduced.
- Analgesia, muscle relaxants, and possibly anesthesia are used to facilitate closed reduction.
- The joint is immobilized by bandages, splints, casts, or traction.
- Neurovascular status is monitored.
- After reduction, if the joint is stable, gentle, progressive, active and passive movement is begun to preserve range of motion (ROM) and restore strength.
- The joint is supported between exercise sessions.

Fractures

- A fracture is a break in the continuity of bone and is defined according to its type and extent.
- Fractures occur when the bone is subjected to stress greater than it can absorb.
- Fractures are caused by direct blows, crushing forces, sudden twisting motions, and extreme muscle contractions.
- When the bone is broken, adjacent structures are also affected, resulting in soft tissue edema, hemorrhage into the muscles and joints, joint dislocations, ruptured tendons, severed nerves, and damaged blood vessels.
- Body organs may be injured by the force that caused the fracture or by fracture fragments.

Types of Fractures

- A complete fracture involves a break across the entire cross-section of the bone and is frequently displaced (removed from its normal position).
- An incomplete fracture (greenstick fracture) involves a break through only part of the cross-section of the bone.
- A comminuted fracture is one that produces several bone fragments.
- A closed fracture (simple) is one that does not cause a break in the skin.
- An open fracture (compound, or complex, fracture) is one in which the skin or mucous membrane wound extends to the fractured bone.
- Avulsion: a fracture in which a fragment of bone has been pulled away by a tendon and its attachment
- Compression: a fracture in which bone has been compressed (seen in vertebral fractures)

- Depressed: a fracture in which fragments are driven inward (seen frequently in fractures of skull and facial bones)
- Epiphyseal: a fracture through the epiphysis
- Greenstick: a fracture in which one side of a bone is broken and the other side is bent
- Impacted: a fracture in which a bone fragment is driven into another bone fragment
- Oblique: a fracture occurring at an angle across the bone (less stable than a transverse fracture)
- Pathologic: a fracture that occurs through an area of diseased bone (eg, osteoporosis, bone cyst, Paget's disease, bony metastasis, tumor); can occur without trauma or a fall
- Spiral: a fracture that twists around the shaft of the bone
- Stress: a fracture that results from repeated loading without bone and muscle recovery
- Transverse: a fracture that is straight across the bone shaft

Clinical Manifestations

- The diagnosis of a fracture is based on the patient's symptoms, the physical signs, and the x-ray findings.
- **Pain**
  - The pain is continuous and increases in severity until the bone fragments are immobilized.
  - The muscle spasms that accompany a fracture begin within 20 minutes after the injury and result in more intense pain than the patient reports at the time of injury.
- **Loss of Function**
  - After a fracture, the extremity cannot function properly because normal function of the muscles depends on the integrity of the bones to which they are attached.
  - Pain contributes to the loss of function.

- **Deformity**
  - Displacement, angulation, or rotation of the fragments in a fracture of the arm or leg causes a deformity (either visible or palpable) that is detectable when the limb is compared with the uninjured extremity.
  - Deformity also results from soft tissue swelling.
- **Shortening**
  - In fractures of long bones, there is actual shortening of the extremity because of the contraction of the muscles that are attached distal and proximal to the site of the fracture.
  - The fragments often overlap by as much as 2.5 to 5 cm (1 to 2 inches).
- **Crepitus**
  - When the extremity is examined with the hands, a grating sensation, called crepitus, can be felt.
  - It is caused by the rubbing of the bone fragments against each other.
- **Swelling and Discoloration**
  - Localized edema and discoloration of the skin (ecchymosis) occur after a fracture as a result of trauma and bleeding into the tissues. These signs may not develop for several hours after the injury.

Emergency Management

- Immobilize the body part before the patient is moved.
- Adequate splinting is essential.
- Joints proximal and distal to the fracture must be immobilized to prevent movement of fracture fragments that can cause additional pain, soft tissue damage, and bleeding.
- Temporary, well-padded splints, firmly bandaged over clothing, serve to immobilize the fracture.
- Immobilization of the lower extremities done by bandaging the legs together.
- In an upper extremity injury, the arm may be bandaged to the chest or may be placed in a sling.

- The neurovascular status distal to the injury should be assessed.
- With an open fracture, the wound is covered with a sterile dressing to prevent contamination.
- In the emergency department, the patient is evaluated completely.
- The clothes are gently removed.
- The patient's clothing may be cut away.

Management

Reduction

- Reduction of a fracture refers to restoration of the fracture fragments to anatomic alignment and rotation.
- The physician reduces a fracture as soon as possible to prevent loss of elasticity from the tissues through infiltration by edema or hemorrhage.
- Fracture reduction becomes more difficult as the injury begins healing.

- **Patient preparation for the procedure:**
  - Permission for the procedure.
  - Analgesic is administered.
  - Anesthesia may be administered.
  - The injured extremity must be handled gently to avoid additional damage.
- **Closed Reduction**
  - Done by bringing the bone fragments into apposition through manipulation and manual traction.
  - The extremity is held in the desired position while the physician applies a cast, splint, or other device.
  - Reduction under anesthesia with percutaneous pinning may be used.
  - Traction (skin or skeletal) may be used to effect fracture reduction and immobilization.
- **Open Reduction**
  - Through a surgical approach, the fracture fragments are reduced.
  - Internal fixation devices (metallic pins, wires, screws) may be used to hold the bone fragments in position until solid bone healing occurs.

Immobilization

- After the fracture has been reduced, the bone fragments must be immobilized.
- Immobilization may be accomplished by external or internal fixation.
- Methods of external fixation include bandages, casts, splints, continuous traction, and external fixators.
- Metal implants used for internal fixation serve as internal splints to immobilize the fracture.

Maintaining and Restoring Function

- Edema is controlled by elevating the injured extremity and applying ice.
- Neurovascular status is monitored.
- Restlessness, anxiety, and discomfort are controlled with reassurance, position changes, and pain relief strategies, including use of analgesics.
- Isometric and muscle-setting exercises are encouraged to minimize disuse atrophy and to promote circulation.
- Participation in activities of daily living (ADLs) is encouraged to promote independent functioning and self-esteem.
- Gradual resumption of activities is promoted.
- With internal fixation, the surgeon determines the amount of movement and weight-bearing stress the extremity can withstand and prescribes the level of activity.

Factors That Enhance Fracture Healing

- Immobilization of fracture fragments
- Maximum bone fragment contact
- Sufficient blood supply
- Proper nutrition
- Exercise: weight bearing for long bones

- Hormones: growth hormone, thyroid, calcitonin, vitamin D, anabolic steroids
- Electric potential across fracture

Factors That Inhibit Fracture Healing

- Extensive local trauma
- Bone loss
- Inadequate immobilization
- Space or tissue between bone fragments
- Infection
- Local malignancy
- Metabolic bone disease (eg, Paget's disease)
- Irradiated bone (radiation necrosis)
- Avascular necrosis
- Intra-articular fracture (synovial fluid contains fibrolysins, which lyse the initial clot and retard clot formation)
- Age (elderly persons heal more slowly)
- Corticosteroids (inhibit the repair rate)

Fracture Healing and Complications

- **Early complications**
  - Shock
  - Fat embolism
  - Compartment syndrome
  - Deep vein thrombosis [DVT]
  - Pulmonary embolism [PE]
  - Disseminated intravascular coagulation
  - Infection.

- **Delayed complications**
  - Delayed union
  - Malunion
  - Nonunion
  - AVN of bone
  - Reaction to internal fixation devices
  - Complex regional pain syndrome
  - Heterotopic ossification.

Early Complications

Shock

- Hypovolemic shock resulting from hemorrhage from loss of intravascular volume into the interstitial space.
- Because the bone is very vascular, large quantities of blood may be lost as a result of trauma, especially in fractures of the femur and pelvis.
- **Treatment** of shock consists of stabilizing the fracture to prevent further hemorrhage, restoring blood volume and circulation, relieving the patient's pain, providing adequate splinting.

Fat Embolism Syndrome

- At the time of fracture, fat globules may diffuse into the vascular compartment because the marrow pressure is greater than the capillary pressure or because catecholamines elevated by the patient's stress reaction mobilize fatty acids and promote the development of fat globules in the bloodstream.
- The fat globules (emboli) may occlude the small blood vessels that supply the lungs, brain, kidneys, and other organs.
- The onset of symptoms is rapid, usually within 24 to 72 hours of injury, but may occur up to a week after injury.

Clinical Manifestations

- hypoxia, tachycardia, and pyrexia, tachypnea, dyspnea, crackles, wheezes, chest pain, cough, large amounts of thick white sputum.
- The chest x-ray shows a typical “snowstorm” infiltrate.
- Headache and mild agitation to delirium and coma.

Prevention

- Immediate immobilization of fractures
- Minimal fracture manipulation
- Adequate support for fractured bones during turning and positioning
- Maintenance of fluid and electrolyte balance.
- The nurse monitors high-risk patients (adults between 20 and 30 years of age with long bone, pelvic, or multiple fractures or crush injuries, and elderly patients with hip fractures).

Compartment Syndrome

- Compartment syndrome is a complication that develops when pressure within a compartment is greater than normal.
- A hallmark sign is pain that occurs or intensifies with passive ROM.
- This pain can be caused by
- A reduction in the size of the muscle compartment because the enclosing muscle fascia is too tight or a cast or dressing is constrictive
- An increase in compartment contents because of edema or hemorrhage (eg, fractures, venomous snake bites, burns).
- The lower leg is most frequently involved, but the forearm is also at risk.
- The pressure within a muscle compartment may increase to such an extent as to decrease microcirculation, causing nerve and muscle anoxia and necrosis.
- Permanent function can be lost if the anoxic situation continues for longer than 6 hours.

Assessment and Diagnostic Findings

- Frequent assessment of neurovascular function after a fracture is essential and focuses on the “5 P's:” pain, paralysis, paresthesias, pallor, and pulselessness.
- Paresthesia (burning or tingling sensation) and numbness are early signs of nerve involvement.
- Motion is evaluated by asking the patient to move the fingers or toes distal to the potential problem.
- With continued nerve ischemia and edema, the patient experiences sensations of hypoesthesia (diminished sensitivity to stimulation) and then absence of feeling.
- Motor weakness may occur as a late sign of nerve ischemia.
- No movement (paralysis) suggests nerve damage.
- Doppler ultrasonography may be used to verify a pulse.

Delayed Complications

Delayed Union, Malunion, and Nonunion

- Delayed union occurs when healing does not occur at a normal rate for the location and type of fracture.
- Nonunion results from failure of the ends of a fractured bone to unite.
- Malunion results from failure of the ends of a fractured bone to unite in normal alignment.
- Factors contributing to nonunion and malunion include infection, inadequate immobilization or manipulation, excessive space between bone fragments (bone gap), limited bone contact, and impaired blood supply.
- In nonunion, fibrocartilage or fibrous tissue exists between the bone fragments.

Management

- Internal fixation stabilizes the bone fragments and ensures bone contact.
- Bone grafts provide for osteogenesis, osteoconduction, or osteoinduction.
- Osteogenesis (bone formation) occurs after transplantation of bone containing osteoblasts.
- Osteoconduction is provision by the graft of the structural matrix for ingrowth of blood vessels and osteoblasts.
- Osteoinduction is the stimulation of host stem cells to differentiate into osteoblasts.
- The electrical stimulation modifies the tissue environment, making it electronegative, which enhances mineral deposition and bone formation that promotes bone growth.

Avascular Necrosis of Bone

- Avascular necrosis (AVN) occurs when the bone loses its blood supply and dies.
- It may occur after a fracture with disruption of the blood supply (especially of the femoral neck).
- It is also seen with dislocations, bone transplantation, prolonged high-dose corticosteroid therapy, chronic renal disease, sickle cell anemia, and other diseases.
- X-rays reveal loss of mineralized matrix and structural collapse.
- Treatment generally consists of attempts to revitalize the bone with bone grafts, prosthetic replacement.

Heterotopic Ossification

- Heterotopic ossification is the abnormal formation of bone, near bones or in muscle, in response to soft tissue trauma after blunt trauma, fracture, or total joint replacement.

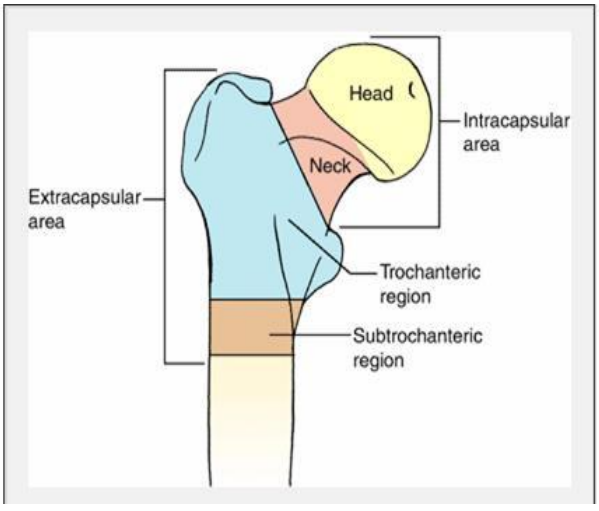
- The muscle is painful, and normal muscular contraction and movement are limited.
- Early mobilization may prevent its occurrence. NSAIDs may be used prophylactically if deep muscle contusion has occurred.
- Usually, the bone lesion resorbs over time, but the abnormal bone eventually may need to be excised if symptoms persist.

Fracture of the Femur

- Elderly people (particularly women) who have brittle bones from osteoporosis and who tend to fall frequently have a high incidence of hip fracture.
- There are two major types of hip fracture.
  - Intracapsular fractures are fractures of the neck of the femur.
  - Extracapsular fractures are fractures of the trochanteric region and of the subtrochanteric region.
- Fractures of the neck of the femur may damage the vascular system that supplies blood to the head and the neck of the femur, and the bone may become ischemic.
- For this reason, AVN is common in patients with femoral neck fractures.
- Extracapsular intertrochanteric fractures have an excellent blood supply and heal readily.
- The elderly are particularly vulnerable to intertrochanteric fractures and generally have a much worse prognosis than younger patients.

Clinical Manifestations

- With fractures of the femoral neck, the leg is shortened, adducted, and externally rotated.
- The patient complains of pain in the hip and groin or in the medial side of the knee.
- With most fractures of the femoral neck, the patient cannot move the leg without a significant increase in pain.
- The patient is most comfortable with the leg slightly flexed in external rotation.



Gerontologic Considerations

- Hip fractures are a frequent contributor to death after 75 years of age.
- Stress and immobility related to the trauma predispose the older adult to atelectasis, pneumonia, sepsis, venous thromboemboli, and reduced ability to cope with other health problems.
- Chronic respiratory problems may be present and may contribute to the possible development of atelectasis or pneumonia.
- Coughing and deep-breathing exercises are encouraged.
- Frequently, elderly people are taking cardiac, antihypertensive, or respiratory medications that need to be continued.
- Dehydration and poor nutrition may be present.
- Dehydration contributes to hemoconcentration and predisposes the patient to the development of venous thromboemboli.
- Therefore, the patient needs to be encouraged to consume adequate fluids and a healthy diet.
- Bed rest and immobility cause an additional loss of muscle strength unless the nurse encourages the patient to move all joints except the involved hip and knee.
- Patients are encouraged to use their arms and the overhead trapeze to reposition themselves.
- This strengthens the arms and shoulders, which facilitates walking with assistive devices.

Medical Management

- Temporary skin traction, may be applied to reduce muscle spasm, to immobilize the extremity, and to relieve pain.
- The goal of surgical treatment of hip fractures is to obtain a fixation so that the patient can be mobilized quickly.

Surgical treatment consists of:

- Open reduction of the fracture and internal fixation.
- Closed reduction with percutaneous stabilization for an intracapsular fracture.
- Replacement of the femoral head with a prosthesis (hemiarthroplasty).
- Surgical intervention is carried out as soon as possible after injury.

Medical Management

- After general or spinal anesthesia, the hip fracture is reduced under x-ray visualization using an image intensifier.
- A stable fracture is usually fixed with nails, a nail-and-plate combination, multiple pins, or compression screw devices.
- Adequate reduction is important for fracture healing: the better the reduction, the better the healing.
- Total hip replacement may be used in selected patients with acetabular defects.

Nursing Management

- Attention is given to pain management, prevention of secondary medical problems, and early mobilization of the patient so that independent functioning can be restored.
- The nurse encourages deep-breathing, coughing, and foot flexion exercises every 1 to 2 hours.
- Thigh-high elastic compression stockings and pneumatic compression devices are used and anticoagulants are administered to prevent the formation of venous thromboemboli.
- The nurse administers prescribed prophylactic IV antibiotics and monitors the patient's hydration, nutritional status, and urine output.

- A pillow is placed between the legs to maintain abduction and alignment and to provide needed support when turning the patient.

Promoting Exercise

- The patient is encouraged to exercise as much as possible by means of the overbed trapeze.
- On the first postoperative day, the patient transfers to a chair with assistance and begins assisted ambulation.
- In general, hip flexion and internal rotation restrictions apply only if the patient has had a hemiarthroplasty.
- Physical therapists work with the patient on transfers, ambulation, and the safe use of a walker and crutches.
- The patient can anticipate discharge to home or to an extended care facility with the use of an ambulatory aid.

Monitoring and Managing Potential Complications

- Neurovascular complications may occur from direct injury to nerves and blood vessels or from increased tissue pressure.
- With hip fracture, bleeding into the tissues is expected.
- DVT is the most common complication.
- To prevent DVT, the nurse encourages intake of fluids and ankle and foot exercises.
- Elastic compression stockings and prophylactic anticoagulant therapy may be prescribed.
- The nurse assesses the patient's legs at least every 4 hours for signs of DVT, which may include unilateral calf tenderness, warmth, redness, and swelling.
- Pulmonary complications, which commonly include atelectasis and pneumonia.

- Deep-breathing exercises, a change of position at least every 2 hours, and the use of an incentive spirometer help prevent respiratory complications.
- Pain must be treated with analgesic agents, typically opioids.
- The nurse assesses breath sounds at least every 4 hours to detect adventitious or diminished sounds.
- Skin breakdown is often seen in elderly patients with hip fracture.
- Patients with hip fractures tend to remain in one position and may develop pressure ulcers.
- Proper skin care helps to relieve pressure.
- Air mattress may provide protection by distributing pressure evenly.

Assessment of Renal and Urinary Tract Function

Identifying Characteristics of Genitourinary Pain

Type	Location	Character	Signs and Symptoms	Etiology
Kidney	Costovertebral angle, may extend to umbilicus	Dull constant ache; if sudden distention of capsule, pain is severe, sharp, stabbing, and colicky in nature	Nausea and vomiting, diaphoresis, pallor, signs of shock	Acute obstruction, kidney stone, blood clot, acute pyelonephritis, trauma
Bladder	Suprapubic area	Dull, continuous pain, may be intense with voiding, may be severe if bladder full	Urgency, pain at end of voiding, painful straining	Overdistended bladder, infection, interstitial cystitis; tumor
Ureteral	Costovertebral angle, flank, lower abdominal area, testis, or labium	Severe, sharp, stabbing pain, colicky in nature	Nausea and vomiting, paralytic ileus	Ureteral stone, edema or stricture, blood clot
Prostatic	Perineum and rectum	Vague discomfort, feeling of fullness in perineum, vague back pain	Suprapubic tenderness, obstruction to urine flow; frequency, urgency, dysuria, nocturia	Prostatic cancer, acute or chronic prostatitis
Urethral	Male: along penis to meatus; female: urethra to meatus	Pain variable, most severe during and immediately after voiding	Frequency, urgency, dysuria, nocturia, urethral discharge	Irritation of bladder neck, infection of urethra, trauma, foreign body in lower urinary tract

Problem	Definition	Possible Etiology
Frequency	Frequent voiding—more than every 3 h	Infection, obstruction of lower urinary tract leading to residual urine and overflow, anxiety, diuretics, benign prostatic hyperplasia, urethral stricture, diabetic neuropathy
Urgency	Strong desire to void	Infection, chronic prostatitis, urethritis, obstruction of lower urinary tract leading to residual urine and overflow, anxiety, diuretics, benign prostatic hyperplasia, urethral stricture, diabetic neuropathy
Dysuria	Painful or difficult voiding	Lower urinary tract infection, inflammation of bladder or urethra, acute prostatitis, stones, foreign bodies, tumors in bladder
Hesitancy	Delay, difficulty in initiating voiding	Benign prostatic hyperplasia, compression of urethra, outlet obstruction, neurogenic bladder
Nocturia	Excessive urination at night	Decreased renal concentrating ability, heart failure, diabetes mellitus, incomplete bladder emptying, excessive fluid intake at bedtime, nephrotic syndrome, cirrhosis with ascites
Incontinence	Involuntary loss of urine	External urinary sphincter injury, obstetric injury, lesions of bladder neck, detrusor dysfunction, infection, neurogenic bladder, medications, neurologic abnormalities
Enuresis	Involuntary voiding during sleep	Delay in functional maturation of central nervous system (bladder control usually achieved by 5 y of age), obstructive disease of lower urinary tract, genetic factors, failure to concentrate urine, urinary tract infection, psychological stress
Polyuria	Increased volume of urine voided more than 3000 mL/d	Diabetes mellitus, diabetes insipidus, use of diuretics, excess fluid intake, lithium toxicity, some forms of kidney disease (hypercalcemic and hypokalemic nephropathy)

Oliguria	Urine output less than 400 mL/d	Acute or chronic renal failure, inadequate fluid intake
Anuria	Urine output less than 50 mL/d	Acute or chronic renal failure, complete obstruction
Hematuria	Red blood cells in the urine	Cancer of genitourinary tract, acute glomerulonephritis, renal stones, renal tuberculosis, Polyuria , trauma, extreme exercise, rheumatic fever, hemophilia, leukemia, sickle cell trait or disease
Proteinuria	Abnormal amounts of protein in the urine	Acute and chronic renal disease, nephrotic syndrome, vigorous exercise, heat stroke, severe heart failure, diabetic nephropathy, multiple myeloma

Problems Associated With Changes in Voiding

Diagnostic Evaluation

Urinalysis and Urine Culture

- The urinalysis provides important clinical information about kidney function and helps diagnose other diseases, such as diabetes.
- The urine culture determines whether bacteria are present in the urine, as well as their strains and concentration.
- Urine culture and sensitivity also identify the antimicrobial therapy that is best suited for the particular strains identified.
- Urine examination includes the following:
    - Urine color
    - Urine clarity and odor
    - Urine pH and specific gravity
    - Tests to detect protein, glucose, and ketone bodies in the urine (proteinuria, glycosuria, and ketonuria, respectively)
    - Microscopic examination of the urine sediment after centrifugation to detect RBCs (hematuria), white blood cells, casts (cylindruria), crystals (crystalluria), pus (pyuria), and bacteria (bacteriuria)

Urine Color	**Possible Cause**
Colorless to pale yellow	Dilute urine due to diuretics, alcohol consumption, diabetes insipidus, glycosuria, excess fluid intake, renal disease
Yellow to milky white	Pyuria, infection
Bright yellow	Multiple vitamin preparations
Pink to red	Hemoglobin breakdown, red blood cells, gross blood, menses, bladder or prostate surgery, medications (phenytoin, rifampin)
Blue, blue green	Dyes, *Pseudomonas* species organisms
Orange to amber	Concentrated urine due to dehydration, fever, bile, excess bilirubin, medications (phenazopyridium HCl)
Brown to black	Old red blood cells, urobilinogen, extremely concentrated urine due to dehydration, medications (metronidazole, iron preparations)

<table>
<tr><th>Test</th><th>Purpose</th><th>Normal Values</th></tr>
<tr><td colspan="3">Renal Concentration Tests</td></tr>
<tr><td>Specific gravity</td><td>Evaluates ability of kidneys to concentrate solutes in urine.</td><td>1.010 to 1.025</td></tr>
<tr><td>Urine osmolality</td><td>Concentrating ability is lost early in kidney disease; hence, these test findings may disclose early defects in renal function.</td><td>300–900 mOsm/kg/24 h, 50–1200 mOsm/kg random sample</td></tr>
<tr><td colspan="3">24-Hour Urine Test</td></tr>
<tr><td>Creatinine clearance</td><td>Detects and evaluates progression of renal disease. Test measures volume of blood cleared of endogenous creatinine in 1 min, which provides an approximation of the glomerular filtration rate. Sensitive indicator of renal disease used to follow progression of renal disease.</td><td>Measured in mL/min/1.73 m^2
<table>
<tr><th>Age</th><th>Male</th><th>Female</th></tr>
<tr><td>Under 30</td><td>88–146</td><td>81–134</td></tr>
<tr><td>30–40</td><td>82–140</td><td>75–128</td></tr>
<tr><td>40–50</td><td>75–133</td><td>69–122</td></tr>
<tr><td>50–60</td><td>68–126</td><td>64–116</td></tr>
<tr><td>60–70</td><td>61–120</td><td>58–110</td></tr>
<tr><td>70–80</td><td>55–113</td><td>52–105</td></tr>
</table></td></tr>
<tr><td colspan="3">Serum Tests</td></tr>
<tr><td>Creatinine level</td><td>Measures effectiveness of renal function. Creatinine is end product of muscle energy metabolism. In normal function, level of creatinine, which is regulated and excreted by the kidneys, remains fairly constant in body.</td><td>0.6–1.2 mg/dL (50–110 mmol/L)</td></tr>
<tr><td>Urea nitrogen (blood urea nitrogen [BUN])</td><td>Serves as index of renal function. Urea is nitrogenous end product of protein metabolism. Test values are affected by protein intake, tissue breakdown, and fluid volume changes.</td><td>7–18 mg/dL
Patients > 60 y: 8–20 mg/dL</td></tr>
<tr><td>BUN to creatinine ratio</td><td>Evaluates hydration status. An elevated ratio is seen in hypovolemia; a normal ratio with an elevated BUN and creatinine is seen with intrinsic renal disease.</td><td>About 10:1</td></tr>
</table>

Renal Function Tests

Imaging Modalities

Kidney, Ureter, and Bladder Studies

An x-ray study of the abdomen or kidneys, ureters, and bladder (KUB) may be performed to delineate the size, shape, and position of the kidneys and to reveal any abnormalities, such as calculi (stones) in the kidneys or urinary tract, hydronephrosis (distention of the pelvis of the kidney), cysts, tumors, or kidney displacement by abnormalities in surrounding tissues.

General Ultrasonography

Ultrasonography is a noninvasive procedure. Abnormalities such as fluid accumulation, masses, congenital malformations, changes in organ size, and obstructions can be identified. During the test, the lower abdomen and genitalia may need to be exposed. Ultrasonography requires a full bladder; therefore, fluid intake should be encouraged before the procedure.

Bladder Ultrasonography

Bladder ultrasonography is a noninvasive method of measuring urine volume in the bladder. It may be indicated for urinary frequency, inability to void after removal of an indwelling urinary catheter, measurement of postvoiding residual urine volume, inability to void postoperatively, or assessment of the need for catheterization during the initial stages of an intermittent catheterization training program.

Computed Tomography and Magnetic Resonance Imaging

They are used to evaluate genitourinary masses, nephrolithiasis, chronic renal infections, renal or urinary tract trauma, metastatic disease, and soft tissue abnormalities.

Oral or IV radiopaque contrast agent is used in CT scanning to enhance visualization.

Nuclear Scans

Nuclear scans are used to evaluate acute and chronic renal failure, renal masses, and blood flow before and after kidney transplantation. The radioisotope is injected at a specified time before the study to achieve the proper concentration in the kidneys. After the procedure is completed, the patient is encouraged to drink fluids to promote excretion of the radioisotope by the kidneys.

Intravenous Urography

IV urography includes various tests such as excretory urography, intravenous pyelography (IVP), and infusion drip pyelography. A radiopaque contrast agent is administered intravenously. An IVP shows the kidneys, ureter, and bladder via x-ray imaging as the dye moves through the upper and then the lower urinary system. A nephrotomogram may be carried out as part of the study to visualize different layers of the kidney and the diffuse structures within each layer and to differentiate solid masses or lesions from cysts in the kidneys or urinary tract.

IV urography may be used as the initial assessment of many suspected urologic problems, especially lesions in the kidneys and ureters. It also provides an approximate estimate of renal function. After the contrast agent is administered intravenously, multiple x-rays are obtained to visualize drainage structures in the upper and lower urinary systems.

Retrograde Pyelography

In retrograde pyelography, catheters are advanced through the ureters into the renal pelvis by means of cystoscopy. A contrast agent is then injected. Retrograde pyelography is usually performed if IV urography provides inadequate visualization of the collecting systems. Possible complications include infection, hematuria, and perforation of the ureter.

Cystography

Cystography aids in evaluating vesicoureteral reflux (backflow of urine from the bladder into one or both ureters) and in assessing for bladder injury. A catheter is inserted into the bladder, and a contrast agent is instilled to outline the bladder wall. The contrast agent may leak through a small bladder perforation stemming from bladder injury, but such leakage is usually harmless.

Renal Angiography

A renal angiogram, or renal arteriogram, provides an image of the renal arteries. The femoral (or axillary) artery is pierced with a needle, and a catheter is threaded up through the femoral and iliac arteries into the aorta or renal artery. A contrast agent is injected to opacify the renal arterial supply. Angiography is used to evaluate renal blood flow in suspected renal trauma, to differentiate renal cysts from tumors, and to evaluate hypertension. It is used preoperatively for renal transplantation. Before the procedure, a laxative may be prescribed to evacuate the colon so that unobstructed x-rays can be obtained. Injection sites (groin for femoral approach or axilla for axillary approach) may be shaved. The peripheral pulse sites (radial, femoral, and dorsalis pedis) are marked for easy access during postprocedural assessment. The patient is informed that there may be a brief sensation of heat along the course of the vessel when the contrast agent is injected.

After the procedure, vital signs are monitored until stable. If the axillary artery was the injection site, blood pressure measurements are taken on the opposite arm. The injection site is examined for swelling and hematoma. Peripheral pulses are palpated, and the color and temperature of the involved extremity are noted and compared with those of the uninvolved extremity. Cold compresses may be applied to the injection site to decrease edema and pain. Possible complications include hematoma formation, arterial thrombosis or dissection, false aneurysm formation, and altered renal function.

Urologic Endoscopic Procedures

Endourology, or urologic endoscopic procedures, can be performed in one of two ways: using a cystoscope inserted into the urethra, or percutaneously, through a small incision.

The cystoscopic examination is used to directly visualize the urethra and bladder. The cystoscope, which is inserted through the urethra into the bladder, has a self-contained optical lens system that provides a magnified, illuminated view of the bladder. The use of a high-intensity light and interchangeable lenses allows excellent visualization and permits still and motion pictures to be taken. The cystoscope is manipulated to allow complete visualization of the urethra and bladder as well as the ureteral orifices and prostatic urethra. Small ureteral catheters can be passed through the cystoscope, allowing assessment of the ureters and the pelvis of each kidney.

The cystoscope also permits the urologist to obtain a urine specimen from each kidney to evaluate its function. Cup forceps can be inserted through the cystoscope for biopsy. Calculi may be removed from the urethra, bladder, and ureter using cystoscopy. If a lower tract cystoscopy is performed, the patient is usually conscious, and the procedure is usually no more uncomfortable than a catheterization.

To minimize post-test urethral discomfort, viscous lidocaine is administered several minutes before the study. If the cystoscopy includes examination of the upper tracts, a sedative may be administered before the procedure. General anesthesia is usually administered to ensure that there are no involuntary muscle spasms when the scope is being passed through the ureters or kidney.

The nurse describes the procedure to the patient and family to prepare them and to allay their fears. If an upper cystoscopy is to be performed, the patient is usually restricted to nothing by mouth (NPO) for several hours beforehand.

Postprocedural management is directed at relieving any discomfort resulting from the examination. Some burning on voiding, blood-tinged urine, and urinary frequency from trauma to the mucous membranes can be expected. Moist heat to the lower abdomen and warm sitz baths are helpful in relieving pain and relaxing the muscles.

After a cystoscopic examination, the patient with obstructive pathology may experience urine retention if the instruments used during the examination caused edema. The nurse carefully monitors the patient with prostatic hyperplasia for urine retention. Warm sitz baths and antispasmodic medication, such as flavoxate (Urispas), may be prescribed to relieve temporary urine retention caused by poor relaxation of the urinary sphincter; however, intermittent catheterization may be necessary for a few hours after the examination. The nurse monitors the patient for signs and symptoms of urinary tract infection. Because edema of the urethra secondary to local trauma may obstruct urine flow, the patient is also monitored for signs and symptoms of obstruction.

Kidney Biopsy

Biopsy of the kidney is used to help diagnose and evaluate the extent of kidney disease. Indications for biopsy include unexplained acute renal failure, persistent proteinuria or hematuria, transplant rejection, and glomerulopathies. A small section of renal cortex is obtained either percutaneously (needle biopsy) or by open biopsy through a small flank incision. Before the biopsy is carried out, coagulation studies are conducted to identify any risk of postbiopsy bleeding. Contraindications to kidney biopsy include bleeding tendencies, uncontrolled hypertension, a solitary kidney, and morbid obesity.

Procedure

The patient may be prescribed a fasting regimen 6 to 8 hours before the test. An IV line is established. A urine specimen is obtained and saved for comparison with the postbiopsy specimen.

If a needle biopsy is to be performed, the patient is instructed to breathe in and hold that breath (to prevent the kidney from moving) while the needle is being inserted. The sedated patient is placed in a prone position with a sandbag under the abdomen. The skin at the biopsy site is infiltrated with a local anesthetic. The biopsy needle is introduced just inside the renal capsule of the outer quadrant of the kidney. The location of the needle may be confirmed by fluoroscopy or by ultrasound, in which case a special probe is used.

With open biopsy, a small incision is made over the kidney, allowing direct visualization. Preparation for an open biopsy is similar to that for any major abdominal surgery.

Nursing Implications

Patients frequently feel discomfort and embarrassment about such a private and personal function as voiding. Voiding in the presence of others can frequently cause guarding, a natural reflex that inhibits voiding due to situational anxiety. Because the outcomes of these studies determine the plan of care, the nurse must help the patient relax by providing as much privacy and explanation about the procedure as possible.

Management of Patients with Renal Disorders

Acute Nephritic Syndrome

- The **acute nephritic syndrome** is the clinical manifestation of glomerular inflammation
- **Glomerulonephritis** is an inflammation of the glomerular capillaries that can occur in acute and chronic forms.

Pathophysiology

- Postinfectious causes are group A betahemolytic streptococcal infection of the throat that precedes the onset of glomerulonephritis by 2 to 3 weeks
- It may also follow impetigo (infection of the skin) and acute viral infections (upper respiratory tract infections, mumps, varicella zoster virus, hepatitis B, and human immunodeficiency virus [HIV] infection).
- In some patients, antigens outside the body (eg, medications, foreign serum) initiate the process
- In other patients, the kidney tissue itself serves as the inciting antigen.

Clinical Manifestations

- The primary presenting features of an acute glomerular inflammation are hematuria, edema, **azotemia,** an abnormal concentration of nitrogenous wastes in the blood, and proteinuria.
- The urine may appear cola-colored because of (RBCs).
- Edema and hypertension is present in most patients.
- Marked proteinuria due to the increased permeability of the glomerular membrane may also occur, with associated pitting edema, hypoalbuminemia, hyperlipidemia, and fatty casts in the urine.
- BUN and serum creatinine levels may increase.

- Anemia may be present.
- Headache, malaise, and flank pain.
- Elderly patients may experience circulatory overload with dyspnea, engorged neck veins, cardiomegaly, and pulmonary edema.
- Confusion, somnolence, and seizures.

Assessment and Diagnostic Findings

- In acute nephritic syndrome, the kidneys become large, edematous, and congested.
- Signs of circulatory overload
- Change in urine color and amount
- Mild to moderate hypertension
- Hypoalbuminemia
- Hyperlipidemia
- Rising BUN and creatinine

Complications

- Complications of acute glomerulonephritis include hypertensive encephalopathy, heart failure, and pulmonary edema.
- Hypertensive encephalopathy is a medical emergency, and therapy is directed toward reducing the blood pressure without impairing renal function.
- Rapidly progressive glomerulonephritis is characterized by a rapid decline in renal function.
- ESRD develops in a matter of weeks or months.
- Dialysis is initiated in acute glomerulonephritis if signs and symptoms of uremia are severe.
- The prognosis for patients with acute nephritic syndrome is excellent and rarely causes CKD

Medical Management

- Management consists primarily of treating symptoms, attempting to preserve kidney function, and treating complications promptly.
- Treatment may include corticosteroids, managing hypertension, and controlling proteinuria
- Pharmacologic therapy depends on the cause of acute glomerulonephritis.
- If residual streptococcal infection is suspected, penicillin is the agent of choice
- Dietary protein is restricted when renal insufficiency and elevated BUN
- Sodium is restricted when the patient has hypertension, edema, and heart failure.

Nursing Management

- carbohydrates are given liberally to provide energy and reduce the catabolism of protein.
- I&O are carefully measured and recorded.
- Fluids are given based on the patient's fluid losses and daily body weight.
- Insensible fluid loss through the lungs (300 mL) and skin (600 mL) is considered when estimating fluid loss.
- Proteinuria and microscopic hematuria may persist for many months.
- Patient education about the disease process, explanations of laboratory and other diagnostic tests, and preparation for safe and effective self-care at home.

Nephrotic Syndrome

- **Nephrotic syndrome** is a type of renal failure characterized by increased glomerular permeability and is manifested by massive proteinuria

Clinical findings include:

  - A marked increase in protein (particularly albumin) in the urine (proteinuria)
  - A decrease in albumin in the blood (hypoalbuminemia)
  - Diffuse edema
  - High Serum cholesterol, and low-density lipoproteins (hyperlipidemia)

- Serious damage of the glomerular capillary membrane results in increased glomerular permeability to plasma proteins.
- Although the liver is capable of increasing the production of albumin, it cannot keep up with the daily loss of albumin through the kidneys. Thus, hypoalbuminemia results

Clinical Manifestations

- The major manifestation of nephrotic syndrome is edema.
- It is usually soft and pitting and commonly occurs around the eyes (periorbital), in dependent areas (sacrum, ankles, and hands), and in the abdomen (ascites).
- Irritability, headache, and malaise.

Assessment and Diagnostic Findings

- Proteinuria (predominately albumin) exceeding 3.5 g/day is the hallmark of the diagnosis of nephrotic syndrome
- Protein electrophoresis of the urine to categorize the type of proteinuria.
- The urine may also contain increased white blood cells (WBCs) as well as epithelial casts.

- A needle biopsy of the kidney may be performed for histologic examination of renal tissue to confirm the diagnosis.

Complications

- Infection (due to a deficient immune response)
- Thromboembolism (especially of the renal vein)
- Pulmonary emboli
- ARF (due to hypovolemia)
- Accelerated atherosclerosis (due to hyperlipidemia).

Medical Management

- Treatment is focused on treating the underlying disease state.
- Steroids.
- Typical treatment includes diuretics for edema.
- ACE inhibitors to reduce proteinuria.
- Lipid lowering agents for hyperlipidemia.
- Heparin to reduce coagulability.

Nursing Management

- Nursing management is similar to that of the patient with acute glomerulonephritis.

Acute Renal Failure

- Reversible clinical syndrome whereby there is sudden and pronounced loss of kidney function
- Occurs over hours to days
- life-threatening metabolic complications can occur, including:
- Metabolic acidosis
- Fluid and electrolyte imbalances.
- A widely accepted criterion for ARF is a 50% or greater increase in serum creatinine above baseline.
- Urine volume may be normal, or changes may occur. Possible changes include oliguria (less than 500 mL/day), nonoliguria (greater than 800 mL/day), or anuria (less than 50 mL/day)

Categories of Acute Renal Failure

- **The major categories of ARF are**
  - Prerenal (hypoperfusion of kidney)
  - Intrarenal (actual damage to kidney tissue)
  - Postrenal (obstruction to urine flow)
- Prerenal ARF, which occurs in 60% to 70% of cases, is the result of impaired blood flow that leads to hypoperfusion of the kidney and a decrease in the GFR.
- Intrarenal ARF is the result of actual parenchymal damage to the glomeruli or kidney tubules.
- Acute tubular necrosis (ATN) is the most common type of intrinsic ARF.
- Postrenal ARF usually results from obstruction distal to the kidney. Pressure rises in the kidney tubules and eventually, the GFR decreases.

Prerenal Failure

- **Volume depletion resulting from:**
  - Hemorrhage
  - Renal losses (diuretics, osmotic diuresis)
  - Gastrointestinal losses (vomiting, diarrhea, nasogastric suction)
- **Impaired cardiac efficiency resulting from:**
  - Myocardial infarction
  - Heart failure
  - Dysrhythmias
  - Cardiogenic shock
- **Vasodilation resulting from:**
  - Sepsis
  - Anaphylaxis
  - Antihypertensive medications or other medications that cause vasodilation

Intrarenal Failure

- **Prolonged renal ischemia resulting from:**
  - Myoglobinuria (trauma, crush injuries, burns)
  - Hemoglobinuria (transfusion reaction, hemolytic anemia)
- **Nephrotoxic agents such as:**
  - Aminoglycoside antibiotics (gentamicin, tobramycin)
  - Radiopaque contrast agents
  - Heavy metals (lead, mercury)
  - Solvents and chemicals (carbon tetrachloride, arsenic)
  - NSAIDs
  - ACE inhibitors

- **Infectious processes such as:**
  - Acute pyelonephritis
  - Acute glomerulonephritis

Postrenal Failure

- **Urinary tract obstruction, including:**
  - Calculi (stones)
  - Tumors
  - Benign prostatic hyperplasia
  - Strictures
  - Blood clots

Phases of Acute Renal Failure

- **Initiation period:** occurs with the insult and ends when oliguria develops
- **Oliguria period:** with urinary output, less than 400ml/24hrs. rising potassium, BUN, Cr. Not responsive to fluid challenges.
- **Diuresis period**: gradual increase in urinary output. Beginning recovery. Renal function gradually improves
- **Recovery period:** may take 3-12 months. May have permanent reduction in functioning of 1%-3%.

Clinical Manifestations

- The patient may appear critically ill and lethargic.
- The skin and mucous membranes are dry from dehydration.
- Central nervous system signs and symptoms include drowsiness, headache, muscle twitching, and seizures.

Assessment and Diagnostic Findings

- Elevated BUN and creatinine
- Sodium retention but may be deceptive due to water retention
- Potassium increased

- Hyperkalemia may lead to dysrhythmias, such as ventricular tachycardia and cardiac arrest.
- Phosphorus increased
- Calcium decreased
- H&H decreased
- Urine output varies from scanty to a normal volume, hematuria may be present, and the urine has a low specific gravity (compared with a normal value of 1.010 to 1.025).
- A renal sonogram or a CT or MRI scan may show evidence of anatomical changes

Preventing Acute Renal Failure

- Provide adequate hydration to patients at risk of dehydration:
  - Surgical patients before, during, and after surgery
  - Patients undergoing intensive diagnostic studies requiring fluid restriction and contrast agents (eg, barium enema, IVP)
  - Cancer pt. or gout and those receiving chemotherapy
- Prevent and treat shock promptly with blood and fluid replacement.
- Monitor CVP and hourly UOP of critically ill patients.
- Treat hypotension promptly.
- Continually assess renal function when appropriate.
- Take precautions to ensure that the appropriate blood is administered to the correct patient in order to avoid severe transfusion reactions.
- Prevent and treat infections promptly
- Pay special attention to wounds, burns, and other precursors of sepsis.
- Remove catheters as soon as possible.
- Monitor dosage, duration of use, and blood levels of all medications metabolized or excreted by the kidneys

Medical Management

- Management includes eliminating the underlying cause
- Prerenal azotemia is treated by optimizing renal perfusion
- IV fluids or transfusions of blood products in prerenal failure
- Postrenal failure is treated by relieving the obstruction.
- Intrarenal azotemia is treated with supportive therapy, with removal of causative agents.
- Fluid balance by daily body weight, CVP, serum and urine concentrations, fluid losses, BP.
- The parenteral and oral intake and the output of urine, gastric drainage, stools, wound drainage, and perspiration are calculated.
- Mannitol, furosemide, or Edecrin to initiate diuresis.
- Hemodialysis
- Peritoneal dialysis
- Continuous renal replacement therapies (CRRTs)

Pharmacologic Therapy

- Low dose dopamine
- For Hyperkalemia Kayexalate orally or by retention enema.
- Kayexalate works by exchanging sodium ions for potassium ions in the intestinal tract.
- Sorbitol may be administered in combination with Kayexalate to induce a diarrhea.
- IV dextrose 50%, insulin, and calcium replacement may be administered to shift potassium back into the cells.
- Ventolin nebulizer can lower plasma potassium concentration.
- Diuretic agents are often used to control fluid volume, but they have not been shown to improve recovery.

- The elevated serum phosphate level may be controlled with phosphate-binding agents.

Nutritional Therapy

- Caloric requirements are met with high-carbohydrate meals
- Foods and fluids containing potassium or phosphorus (eg, bananas, citrus fruits and juices, coffee) are restricted.
- Is a catabolic state and if insufficient intake, patient may lose up to 0.5-1 pounds daily.
- Protein needs: non-dialysis patients need 0.6g/kg of body weight, and dialysis patients will need 1-1.5g/kg.
- Fluid restriction=urine volume plus 500ml

Nursing Management

- Monitor fluid and electrolyte balance
- Reduce metabolic demands
- Promote pulmonary function
- Prevent infection
- Provide skin care
- Provide support

Chronic Renal Failure
(End-stage renal disease)

- Progressive, irreversible deterioration in renal function
- Causation: Diabetes mellitus, hypertension, glomerulonephritis, pyelonephritis, polycystic kidney disease, vascular disorders, exposure to lead, cadmium, mercury, and chromium.

Kidney changes

- Nephrons hypertrophy and work harder until 70-80% of renal function is lost
- Nephrons could only compensate by decreasing water reabsorption thus:
- Hyposthenuria—loss of urine concentrating ability occurs
- Polyuria—increased urine output
- Then isosthenuria—fixed osmolality
- Gradual decline in urinary output

Clinical Manifestations

Neurologic

- Weakness and fatigue
- Confusion
- Inability to concentrate
- Disorientation
- Tremors
- Seizures
- Asterixis
- Restlessness of legs
- Burning of soles of feet
- Behavior changes

Integumentary

- Gray-bronze skin color
- Dry, flaky skin
- Pruritus
- Ecchymosis
- Purpura
- Thin, brittle nails
- Coarse, thinning hair

Cardiovascular

- Hypertension
- Pitting edema (feet, hands, sacrum)
- Periorbital edema
- Pericardial friction rub
- Engorged neck veins
- Pericarditis
- Pericardial effusion
- Pericardial tamponade
- Hyperkalemia
- Hyperlipidemia

Pulmonary

- Crackles
- Thick, tenacious sputum
- Depressed cough reflex
- Pleuritic pain
- Shortness of breath
- Tachypnea
- Kussmaul-type respirations
- Uremic pneumonitis

Gastrointestinal

- Ammonia odor to breath ("uremic fetor")
- Metallic taste
- Mouth ulcerations and bleeding
- Anorexia, nausea, and vomiting
- Hiccups
- Constipation or diarrhea
- Bleeding from gastrointestinal tract

Hematologic

- Anemia
- Thrombocytopenia

Reproductive

- Amenorrhea
- Testicular atrophy
- Infertility
- Decreased libido

Musculoskeletal

- Muscle cramps
- Loss of muscle strength
- Renal osteodystrophy
- Bone pain
- Bone fractures
- Foot drop

Assessment and Diagnostic Findings

- Sodium and Water Retention
- Acidosis
- Anemia
- Calcium and Phosphorus Imbalance

Complications

- **Hyperkalemia** due to decreased excretion, metabolic acidosis, catabolism, and excessive intake (diet, medications, fluids)
- **Pericarditis**, **pericardial effusion**.
- **Hypertension** due to sodium and water retention.
- **Anemia** due to decreased erythropoietin.
- **Bone disease and metastatic and vascular calcifications** due to retention of phosphorus, low serum calcium levels.

Medical Management

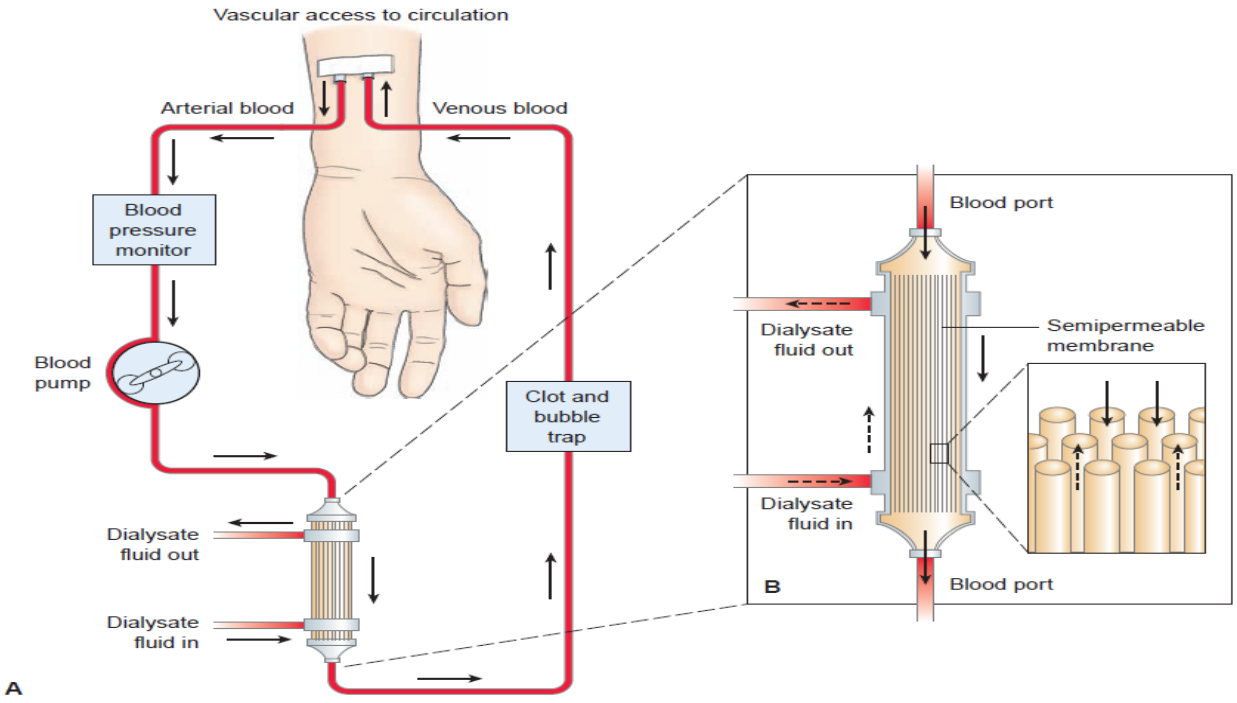
- Calcium and phosphorus binders—Calcium carbonate, calcium acetate
- Antihypertensives and Cardiovascular Agents
- Heart failure and pulmonary edema may also require treatment with fluid restriction, low-sodium diets, diuretic agents, inotropic agents such as digoxin (Lanoxin) or dobutamine (Dobutrex), and dialysis.
- Antiseizure—IV diazepam (Valium) or phenytoin.
- Erythropoietin
- Iron supplements
- Diet—CHO and vitamins, restrict protein, potassium, sodium, fluid 500 mL to 600 mL more than the previous day's 24-hour urine output.

Dialysis

- Types of dialysis include hemodialysis, CRRT, and PD.
- **Acute dialysis is indicated when:**
  - High and increasing level of serum potassium
  - Fluid overload
  - Impending pulmonary edema
  - Increasing acidosis
  - Pericarditis
  - Severe confusion
  - Remove medications or toxins
  - Edema that does not respond to other treatment
  - Hepatic coma, hyperkalemia, hypercalcemia, hypertension, and uremia
- **Chronic or maintenance dialysis** is indicated in advanced CKD and ESRD in the following instances:
  - The presence of uremic signs and symptoms affecting all body systems (nausea and vomiting, severe anorexia, increasing lethargy, mental confusion)
  - Hyperkalemia
  - Fluid overload not responsive to diuretics and fluid restriction
  - General lack of well-being.
- An urgent indication for dialysis in patients with renal failure is pericardial friction rub.

Hemodialysis

- Hemodialysis is the most common method of dialysis.
- Hemodialysis is used for patients who are acutely ill and require short-term dialysis (days to weeks) and for patients with ESRD who require long-term or permanent therapy.
- A dialyzer (also referred to as an artificial kidney) serves as a synthetic semipermeable membrane, replacing the renal glomeruli and tubules as the filter for the impaired kidneys.
- Treatments usually occur three times a week for 3 to 4 hours per treatment.
- The waste products in the blood diffuse through the semipermeable membrane into the dialysate.



Hemodialysis system. A, Blood from an artery is pumped into (B) a dialyzer where it flows through the cellophane tubes, which act as the semipermeable membrane (inset).
The dialysate, which has the same chemical composition as the blood except for urea and waste products, flows in around the tubules.

Principles

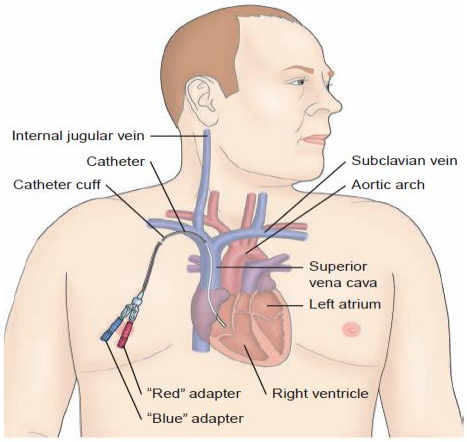
- The toxins and wastes in the blood are removed by **diffusion**—that is, they move from an area of higher concentration in the blood to an area of lower concentration in the dialysate.
- The dialysate is a solution made up of all the important electrolytes in their ideal extracellular concentrations.
- Excess water is removed from the blood by **osmosis,** in which water moves from an area of low concentration potential (the blood) to an area of high concentration potential (the dialysate).
- In **ultrafiltration,** water moves under high pressure to an area of lower pressure.
- This process is much more efficient than osmosis at water removal and is accomplished by applying negative pressure or a suctioning force to the dialysis membrane.

Vascular Access

- Access to the patient's vascular system must be established to allow blood to be removed, cleansed, and returned to the patient's vascular system at rates between 300 and 800 mL/min. Several types of access are available.

Vascular Access Devices

- Immediate access to the patient's circulation for acute hemodialysis is achieved by inserting a double-lumen, noncuffed, large-bore catheter into the subclavian, internal jugular, or femoral vein by the physician



Arteriovenous Fistula

- The preferred method of permanent access is an AVF that is created surgically (usually in the forearm) by joining (anastomosing) an artery to a vein, either side to side or end to side
- AVF will need time, (2 to 3 months) to "mature" before it can be used.
- As the AVF matures, the venous segment dilates due to the increased blood flow coming directly from the artery.
- The patient is encouraged to perform hand exercises to increase the size of these vessels (ie, squeezing a rubber ball for forearm fistulas)
- AVF has the longest useful life and is the best option for chronic hemodialysis patient.

Arteriovenous Graft

- An arteriovenous graft can be created by subcutaneously interposing a biologic, semibiologic, or synthetic graft material between an artery and vein
- Usually a graft is created when the patient's vessels are not suitable for creation of an AV fistula.
- Patients with compromised vascular systems (eg, from diabetes) will require a graft because their vessels are not suitable for creation of an AV Fistula.
- Grafts are usually placed in the arm but may be placed in the thigh or chest area.
- Stenosis, infection, and thrombosis are the most common complications that result in loss of this access.

Complications

- Disturbances of lipid metabolism (hypertriglyceridemia)
- Heart failure, coronary heart disease, angina, stroke, and peripheral vascular insufficiency

- Cardiovascular disease remains the leading cause of death in patients receiving dialysis
- Anemia is compounded by blood lost during hemodialysis.
- Gastric ulcers may result from the physiologic stress
- Vomiting may occur during the hemodialysis treatment when rapid fluid shifts and hypotension occur
- Worsening calcium metabolism and renal osteodystrophy can result in bone pain and fractures
- Calcification of major blood vessels causes hypertension
- Phosphorus deposits in the skin can occur and cause itching
- Sleep problems
- Episodes of shortness of breath
- Painful muscle cramping
- Exsanguination
- Dysrhythmias may result from electrolyte and pH changes
- Air embolism
- Chest pain may occur in patients with anemia
- Dialysis disequilibrium

Dialysis Disequilibrium Syndrome

- Caused by rapid decrease in fluid volume and blood urea nitrogen levels during HD.
- Change in urea levels can cause cerebral edema and increased ICP.
- Neurologic complications include: HA, vomiting, restlessness, decreased LOC, seizures, coma or death.
- Can be prevented by starting HD for short periods and low blood flows.

Nursing Management

- During dialysis, the patient, the dialyzer, and the dialysate require constant monitoring
- Nursing care of the patient and maintenance of the vascular access

Predialysis Nursing Care

- Assess vital signs, including orthostatic blood pressures (lying, sitting, and standing), apical pulse, respirations, and lung sounds.
  - Hypertension may indicate excess fluid volume.
  - Fluid overload may also cause dyspnea, tachypnea, and rales or crackles in the lungs.
  - The client who is hypotensive may not tolerate rapid fluid volume changes during dialysis.
- Record weight.
  - Weight changes are an effective indicator of fluid volume.
- Assess vascular access site for a palpable pulsation or vibration and an audible bruit and for inflammation.
  - Infection and thrombus formation are the most common problems affecting the access site in hemodialysis clients.
- Alert all personnel to avoid using the extremity with the vascular access site for blood pressures or venipuncture.

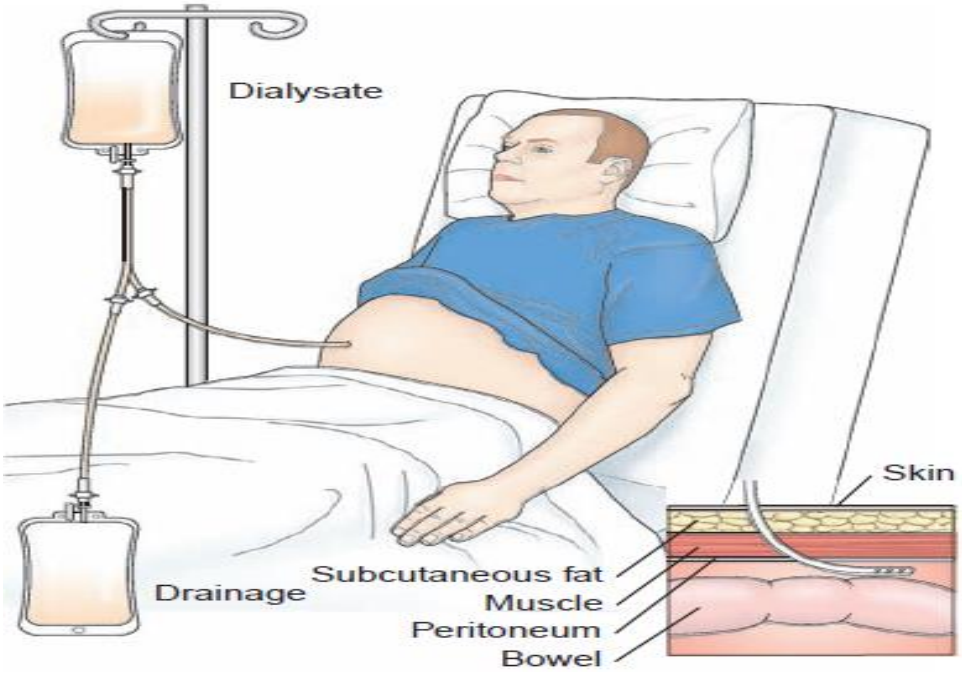
During & Postdialysis Nursing Care

- Assess and document vital signs, weight, and vascular access site condition.
  - Rapid fluid and solute removal during dialysis may lead to orthostatic hypotension, cardiopulmonary changes, and weight loss.
- Monitor BUN, serum creatinine, serum electrolyte, and hematocrit levels between dialysis treatments.
  - These values help determine the effectiveness of the treatment, the need for fluid and diet restrictions, and the timing of future dialysis sessions.
- Iron and folate supplements or periodic blood transfusions may be needed.

- Assess for dialysis disequilibrium syndrome, with headache, nausea and vomiting, altered level of consciousness; and hypertension.
- Assess for other adverse responses to dialysis, such as dehydration, nausea and vomiting, muscle cramps, or seizure activity.
- Assess for bleeding at the access site or elsewhere.
  - Renal failure and heparinization during dialysis increase the risk for bleeding.
- Use standard precautions at all times.
  - Frequent exposure to blood and blood products increase the risk for hepatitis B or C or other bloodborne diseases.
- If a transfusion is given during dialysis, monitor for possible transfusion reaction (e.g., chills and fever; dyspnea; chest, back, or arm pain; and urticaria or itching).
- Provide psychologic support and listen actively.
- Address concerns and accept responses such as anger, depression, and noncompliance.
- Reinforce client and family strengths in coping with renal failure and hemodialysis.
- Refer to social services and counseling as indicated.

Peritoneal Dialysis

- Procedure involves siliconized rubber catheter placed into the abdominal cavity for infusion of dialysate.



Indications

- Patients unwilling to undergo hemodialysis or renal transplantation.
- Patients who are susceptible to the rapid fluid, electrolyte during hemodialysis experience fewer of these problems with the slower rate of peritoneal dialysis.
- Patients with diabetes or cardiovascular disease, many older patients, and those who may be at risk for adverse effects of systemic heparin are likely candidates for peritoneal dialysis.
- Severe hypertension, heart failure, and pulmonary edema.

Complications

- Peritonitis
- Bleeding
- Pain
- Exit site and tunnel infections
- Poor dialysate flow
- Dialysate leakage
- Hypertriglyceridemia
- Abdominal hernias (incisional, inguinal, diaphragmatic, and umbilical)

Nursing Care during Peritoneal Dialysis

- Before treating, evaluate baseline vital signs, weight, and laboratory tests.
- Continually monitor the client for respiratory distress, pain, and discomfort.
- Monitor prescribed dwell time and initiate outflow.
- Observe the outflow amount and pattern of fluid.

Nursing Management of Hospitalized Client on Dialysis

- Protect vascular access
- Monitor fluid balance indicators
- Monitor IV carefully
- Assess for s/s uremia
- Monitor cardiopulmonary status carefully
- Monitor BP
- Monitor medications
- Address pain and discomfort
- Infection control measures
- Monitor dietary electrolytes and fluids
- Skin care

Management of Patients with Urinary Disorders

Infections of the Urinary Tract

Classifying Urinary Tract Infections

- Urinary tract infections (UTIs) are classified by location: the lower urinary tract (which includes the bladder and structures below the bladder) or the upper urinary tract (which includes the kidneys and ureters). They can also be classified as uncomplicated or complicated UTIs.
- **Lower UTI**
  - Cystitis, prostatitis, urethritis
- **Upper UTIs**
  - Acute pyelonephritis, chronic pyelonephritis, renal abscess, interstitial nephritis, perirenal abscess
- **Uncomplicated Lower or Upper UTIs**
  - Community-acquired infection; common in young women and not usually recurrent
- **Complicated Lower or Upper UTIs**
  - Often nosocomial (acquired in the hospital) and related to catheterization; occur in patients with urologic abnormalities, pregnancy, immunosuppression, diabetes mellitus, and obstructions and are often recurrent

Risk Factors for Urinary Tract Infection

- Inability or failure to empty the bladder completely
- Obstructed urinary flow
- Congenital abnormalities
- Urethral strictures

- Contracture of the bladder neck
- Bladder tumors
- Calculi (stones) in the ureters or kidneys
- Compression of the ureters
- Neurologic abnormalities
- Decreased natural host defenses or immunosuppression
- Instrumentation of the urinary tract (eg, catheterization, cystoscopic procedures)
- Inflammation or abrasion of the urethral mucosa
- Diabetes mellitus (increased urinary glucose levels create an infection-prone environment in the urinary tract)
- Pregnancy
- Neurologic disorders
- Gout
- Altered states caused by incomplete emptying of the bladder and urinary stasis

Factors That Contribute to Urinary Tract Infection in Older Adults

- High incidence of chronic illness
- Frequent use of antimicrobial agents
- Presence of infected pressure ulcers
- Immunocompromise
- Cognitive impairment
- Immobility and incomplete emptying of bladder
- Use of a bedpan rather than a commode or toilet

Lower Urinary Tract Infections

- **Several mechanisms maintain the sterility of the bladder:**
  - The physical barrier of the urethra
  - Urine flow
  - Ureterovesical junction competence
  - Antibacterial enzymes and antibodies
  - Antiadherent effects mediated by the mucosal cells of the bladder

Bacteriuria

- Bacteriuria is generally defined as more than 10^5 colonies of bacteria per milliliter of urine. Because urine samples are commonly contaminated by the bacteria normally present in the urethral area, a bacterial count exceeding ***10^5 colonies/mL*** of clean-catch midstream urine is the measure that distinguishes true bacteriuria from contamination.
- In men, contamination of the collected urine sample occurs less frequently; hence, bacteriuria can be defined as ***10^4 colonies/mL*** urine.
- E. coli was responsible for 54.7% of UTIs, Pseudomonas and Enterococcus.

Routes of Infection

- **Bacteria enter the urinary tract in three well-recognized ways:**
  - Transurethral route (ascending infection)
  - Bloodstream (hematogenous spread)
  - A fistula from the intestine (direct extension).

Clinical Manifestations

- 50% asymptomatic
- **Uncomplicated:**
  - Dysuria, burning, frequency, urgency, nocturia, incontinence, suprapubic or pelvic pain.

- Hematuria and back pain may occur.

- **Complicated:**
  - Manifestations may range from asymptomatic bacteruria to gram-negative sepsis and shock.

Assessment and Diagnostic Findings

- **Urine Cultures**
  - A colony count of at least 105 colony-forming units (CFU) per milliliter of urine.
- **Cellular Studies**
  - Microscopic hematuria.
  - Pyuria (greater than 4 white blood cells [WBCs] per high-power field).
- **Other Studies**
  - Tests for sexually transmitted diseases (STDs), also referred to as sexually transmitted infections (STIs), may be performed because acute urethritis caused by sexually transmitted organisms (ie, Chlamydia trachomatis, Neisseria gonorrhoeae, herpes simplex).
  - Diagnostic studies such as (CT) and ultrasonography.
  - An intravenous (IV) urogram.

Preventing Recurrent Urinary Tract Infections

- **Hygiene**
  - Shower rather than bathe in tub because bacteria in the bath water may enter the urethra.
  - After each bowel movement, clean the perineum and urethral meatus from front to back.
- **Fluid Intake**
  - Drink liberal amounts of fluids daily to flush out bacteria.

- Avoid coffee, tea, colas, alcohol, and other fluids that are urinary tract irritants.

- **Voiding Habits**
  - Void every 2 to 3 hours during the day and completely empty the bladder.
- **Therapy**
  - Take medication exactly as prescribed.

Medical Management

- Pharmacologic therapy
- Antibacterial agent
- Antispasmotic
- Antifungal
- Phenazopyridine (Pyridium), a urinary analgesic, may be prescribed to relieve the discomfort associated with the infection
- Education

Upper Urinary Tract Infections

Pyelonephritis is a bacterial infection of the renal pelvis, tubules, and interstitial tissue of one or both kidneys.

Causes

- Upward spread of bacteria from the bladder
- Spread from systemic sources reaching the kidney via the bloodstream.
- An incompetent ureterovesical valve
- Obstruction occurring in the urinary tract because static urine provides a good medium for bacterial growth.
- Bladder tumors, strictures, benign prostatic hyperplasia, and urinary stones
- Systemic infections (such as tuberculosis)

Clinical manifestations

- **Acute**
  - Fever, chills, tachycardia, tachypnea
  - Flank, back or loin pain
  - Abdominal discomfort
  - Dysuria, burning, urgency, frequency, nocturia
  - Nausea and vomiting
  - Headache
  - General malaise or fatigue
- **Chronic**
  - Hypertension
  - Inability to conserve sodium
  - Decreased concentrating ability
  - Tendency to develop hyperkalemia and acidosis

Assessment and Diagnostic Findings (Acute)

- An ultrasound study or a CT scan
- An IV pyelogram is rarely indicated
- Radionuclide imaging
- Urine culture and sensitivity tests

Assessment and Diagnostic Findings (chronic)

- IV urogram and measurements of creatinine clearance, blood urea nitrogen, and creatinine levels.

Complications

- **Complications of chronic pyelonephritis include**
  - End-stage renal disease (ESRD) (from progressive loss of nephrons secondary to chronic inflammation and scarring)
  - Hypertension
  - Formation of kidney stones (from chronic infection with urea-splitting organisms).

Medical Management: Acute

- Outpatient therapy
- Antimicrobial therapy
- Follow-up C&S
- Many need extended antibiotic therapy if reoccurrence
- Pregnant women may be hospitalized for 2 or 3 days of parenteral antibiotic therapy.
- Oral antibiotic agents may be prescribed once the patient is afebrile and showing clinical improvement.
- Hydration with oral or parenteral fluids is essential in all patients with UTIs when there is adequate kidney function. Hydration helps facilitate “flushing” of the urinary tract and reduces pain and discomfort.

Medical Management: Chronic

- Long term use of prophylactic antimicrobial therapy
- Careful monitoring of renal function

Nursing Management

- Hospitalized patient
- Carefully monitoring of I&O
- Increase fluids (3-4 L/day)
- Monitor temperature Q 4 hr
- Administer antipyretic as indicated
- Administer antibiotics
- Patient teaching

Urolithiasis and Nephrolithiasis

- Urolithiasis and nephrolithiasis refer to stones (calculi) in the urinary tract and kidney, respectively.
- Affects men more than women.
- Half of patients with a single renal stone have another episode within 5 years.
    - Calcium 75%
    - Uric acid stones (5% to 10% of all stones) may be seen in patients with gout
    - Struvite (ammonium magnesium phosphate) stones account for 15% of urinary calculi
    - Cystine (amino acid) stones (1% to 2% of all stones).

Causes of hypercalcemia (75%)

- Hyperparathyroidism
- Renal tubular acidosis
- Cancers
- Granulomatous diseases (eg, sarcoidosis, tuberculosis), which may cause increased vitamin D production by the granulomatous tissue
- Excessive intake of vitamin D
- Excessive intake of milk and alkali
- Myeloproliferative diseases (leukemia, polycythemia vera, multiple myeloma)

Risk factors

- Polycystic kidney disease
- Horseshoe kidneys
- Chronic strictures
- Medullary sponge disease
- Patients with inflammatory bowel disease

- Ileostomy or bowel resection because these patients absorb more oxalate.
- Antacids, acetazolamide (diamox), vitamin D, laxatives, and high doses of aspirin.

Clinical Manifestations

- Infection (pyelonephritis and UTI with chills, fever, and dysuria) .
- Stones in the renal pelvis may be associated with an intense, deep ache in the costovertebral region.
- Hematuria is often present.
- Pyuria may also be noted.
- Pain originating in the renal area radiates anteriorly and downward toward the bladder.
- Nausea and vomiting.
- Diarrhea and abdominal discomfort.
- The patient has a desire to void, but little urine is passed, and it usually contains blood because of the abrasive action of the stone.

Assessment and Diagnostic Findings

- KUB, Ultrasonography, IVU
- Retrograde pyelography.
- Blood chemistries and a 24-hour urine test for measurement of calcium, uric acid, creatinine, sodium, pH, and total volume
- Dietary and medication histories and family history of renal stones
- Chemical analysis is carried out to determine their composition.

Management

- **The goals of management are to:**
    - Eradicate the stone
    - Determine the stone type
    - Prevent nephron destruction

- Control infection
- Relieve any obstruction

- The immediate objective of treatment of renal or ureteral colic is to relieve the pain until its cause can be eliminated
  - Opioid analgesics
  - Nonsteroidal anti-inflammatory drugs (NSAIDS)
  - Hot baths or moist heat to the flank areas
  - Fluids are encouraged
- **Nutritional Therapy**
  - Drink eight to ten 8-ounce glasses of water daily
  - IV fluids

Preventing Kidney Stones

- Avoid protein intake to 60 g/day.
- A sodium intake of 3–4 g/day.
- Low-calcium diets are not recommended
- Evidence shows that limiting calcium, especially in women, can lead to osteoporosis and does not prevent renal stones.
- Avoid intake of oxalate-containing foods (eg, spinach, strawberries, rhubarb, tea, peanuts, and wheat bran).
- During the day, drink fluids (ideally water) every 1–2 hours.
- Drink two glasses of water at bedtime and an additional glass at each nighttime awakening to prevent urine from becoming too concentrated during the night.
- Avoid excessive sweating and dehydration.
- Treat promptly a urinary tract infection.

Methods of treating renal stones

- During a cystoscopy, which is used for removing small stones located in the ureter close to the bladder, a ureteroscope is inserted into the ureter to visualize the stone.
  - The stone is then fragmented or captured and removed.
- Extracorporeal shock wave lithotripsy (ESWL) is used for most symptomatic, nonpassable upper urinary tract stones. Electromagnetically generated shock waves are focused over the area of the renal stone.
- The high-energy dry shock waves pass through the skin and fragment the stone.
- Percutaneous nephrolithotomy is used to treat larger stones.
- A percutaneous tract is formed and a nephroscope is inserted through it. Then the stone is extracted or pulverized.
- Chemolysis, stone dissolution using infusions of chemical solutions (eg, alkylating agents, acidifying agents) for the purpose of dissolving the stone.

Surgical Management

- Today, surgery is performed in only 1% to 2% of patients.
- To correct anatomic abnormalities within the kidney to improve urinary drainage.
- If the stone is in the kidney, the surgery performed may be a nephrolithotomy (incision into the kidney with removal of the stone).
- A nephrectomy, if the kidney is nonfunctional.
- Stones in the kidney pelvis are removed by a pyelolithotomy, those in the ureter by ureterolithotomy, and those in the bladder by cystotomy.

Management of Patients with Prostate disorders

Prostatitis

- Prostatitis is an inflammation of the prostate gland that is often associated with lower urinary tract symptoms.
- The condition affects 5% to 10% of men.
- It is the most common urologic diagnosis in men younger than 50 years of age and the third most common such diagnosis in men older than 50 years
- Prostatitis may be caused by infectious agents (bacteria, fungi) or other conditions (eg, urethral stricture, benign prostatic hyperplasia).
- E. coli is the most common causative organism, followed by Klebsiella and Proteus
- There are four types of prostatitis:
  - Acute bacterial prostatitis (type I)
  - Chronic bacterial prostatitis (type II)
  - Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS) (type III)
  - Asymptomatic inflammatory prostatitis (type IV).
- Type III, occurs in more than 90% of cases

Clinical Manifestations

- Acute prostatitis is characterized by the sudden onset of fever, dysuria, perineal prostatic pain and severe lower urinary tract symptoms: dysuria, frequency, urgency, hesitancy, and nocturia.
- Approx. 5% of cases of acute prostatitis progress to chronic bacterial prostatitis.
- Patients with type II disease are asymptomatic between episodes.
- Patients with type III prostatitis often have no bacteria in the urine in the presence of GU pain.
- Patients with type IV prostatitis are usually diagnosed incidentally during a workup for infertility, an elevated PSA test, or other disorders.

Medical Management

- Hospital admission may be necessary for patients with unstable vital signs, sepsis, or intractable pelvic pain; those who are immunosuppressed; or those who have diabetes or renal insufficiency.
- Specific treatment is based on the type of prostatitis and on the results of culture and sensitivity testing of the urine.
- If bacteria are cultured from the urine, antibiotics, including sulprim or ciprofloxacin, may be prescribed.
- If the patient is afebrile and has a normal urinalysis, anti-inflammatory agents may be used.
- Alpha-adrenergic blocker therapy (promote bladder and prostate relaxation).
- Pelvic floor training, physical therapy, sitz baths, stool softeners.

Benign Prostatic Hyperplasia

- In approximately 50% of men 50 years and older, and 80% of men 80 years of age and older.
- The prostate gland enlarges, extending upward into the bladder and obstructing the outflow of urine by encroaching on the vesical orifice.

Risk Factors

Primary risk factors for BPH or enlarged prostate include:

- Aging — the main risk factor for the BPH or enlarged prostate
- Heredity — a family history of BPH or enlarged prostate
- Marital status — for reasons that are not known, men who are married are more likely to develop BPH or enlarged prostate than single men
- Nationality — BPH or enlarged prostate is more common in Americans and Europeans than in Asian men.

Clinical Manifestations

- Frequency of urination, nocturia, urgency, hesitancy in starting urination, abdominal straining with urination, a decrease in the volume and force of the urinary stream, interruption of the urinary stream, dribbling (urine dribbles out after urination), a sensation that the bladder has not been completely emptied, acute urinary retention (more than 60 mL of urine remaining in the bladder after urination), and recurrent urinary tract infections.
- Azotemia (accumulation of nitrogenous waste products) and renal failure can occur with chronic urinary retention and large residual volumes.
- Generalized symptoms, including fatigue, anorexia, nausea, vomiting, and epigastric discomfort.

Assessment and Diagnostic Findings

- DRE: a large, rubbery, and nontender prostate gland.
- Urinalysis and urodynamic studies to assess urine flow.
- Renal function tests.
- Complete blood studies are performed.
- Because hemorrhage is a major complication of prostate surgery, all clotting defects must be corrected.
- Cardiac and respiratory function is also assessed.

Management

- If the patient is admitted on an emergency basis because he cannot void, he is immediately catheterized.
- The ordinary catheter may be too soft and pliable to advance through the urethra into the bladder.
- In such cases, a thin wire called a stylet is introduced (by a urologist) into the catheter to prevent the catheter from collapsing when it encounters resistance.
- In severe cases, metal catheters with a pronounced prostatic curve may be used.

- Sometimes an incision is made into the bladder (a suprapubic cystostomy) to provide drainage.

Pharmacologic Therapy

- Alpha-adrenergic blockers: relax the smooth muscle of the bladder neck and prostate, improves urine flow and relieves BPH symptoms.
- Because a hormonal component of BPH has been identified, one method of treatment involves hormonal manipulation with antiandrogen agents.
    - Side effects include gynecomastia (breast enlargement), and flushing.

Cancer of the Prostate

Risk Factors for Prostate Cancer

- Gender - Only males have prostates
- Age - Most commonly strikes men 55 and older
- Heredity - A family history of prostate cancer increases the risk, especially when the disease is diagnosed in men younger than 50 years of age.
- Race - African Americans are at the highest risk and Asian Americans at the lowest
- High-fat diet: excessive amounts of red meat or dairy products that are high in fat
- Sedentary lifestyle
- Smoking
- Workplace exposure to heavy metals such as cadmium
- Vasectomy
- Infection
    - Researchers have also looked to see if sexually transmitted infections (like gonorrhea or chlamydia) might increase the risk of prostate cancer
- Inflammation of the prostate

Clinical Manifestations

- Most early-stage prostate cancers are asymptomatic.
- Symptoms due to obstruction of urinary flow:
    - Hesitancy and straining on voiding, frequency, nocturia
    - Diminution in size and force of urinary stream
- Symptoms due to metastasis:
    - Pain in lumbosacral area radiating to hips and down legs (from bone metastases)
    - Perineal and rectal discomfort
    - Anemia, weight loss, weakness, nausea, oliguria (from uremia)

- Hematuria (from urethral or bladder invasion, or both)
- Lower extremity edema occurs when pelvic node metastases compromise venous return

Diagnostic Evaluation

- Digital rectal examination prostate can be felt through the wall of the rectum; hard nodule may be felt.
- Needle biopsy (through anterior rectal wall or through perineum) for histologic study of biopsied tissue, includes Gleason tumor grade if carcinoma present.
- Transrectal ultrasonography a sonar probe placed in rectum.
- PSA serologic marker of prostate cancer.
- X-rays, CT scan, MRI, bone scan, analysis of pelvic lymph nodes.

Management

Conservative Measures

- No treatment may be indicated in men over age 70 because prostate cancer may be slow growing and it is expected that many men will die from other causes.
- It is commonly recommended that these patients be followed closely with periodic PSA determinations and examination for evidence of metastases.
- Symptom control for advanced prostatic cancer in which treatment is not effective:
  - Analgesics and opioids to relieve pain.
  - Short course of radiotherapy for specific sites of bone pain.
  - TURP to remove obstructing tissue if bladder outlet obstruction occurs.
  - Suprapubic catheter placement.

Surgical Interventions (Curative)

- Radical prostatectomy removal of entire prostate gland, prostatic capsule, and seminal vesicles; may include pelvic lymphadenectomy.
  - Procedure is used to treat stage A and B prostate cancers.
  - Complications include urinary incontinence and impotence, possible rectal injury.
  - Newer nerve-sparing techniques may preserve sexual potency and continence.

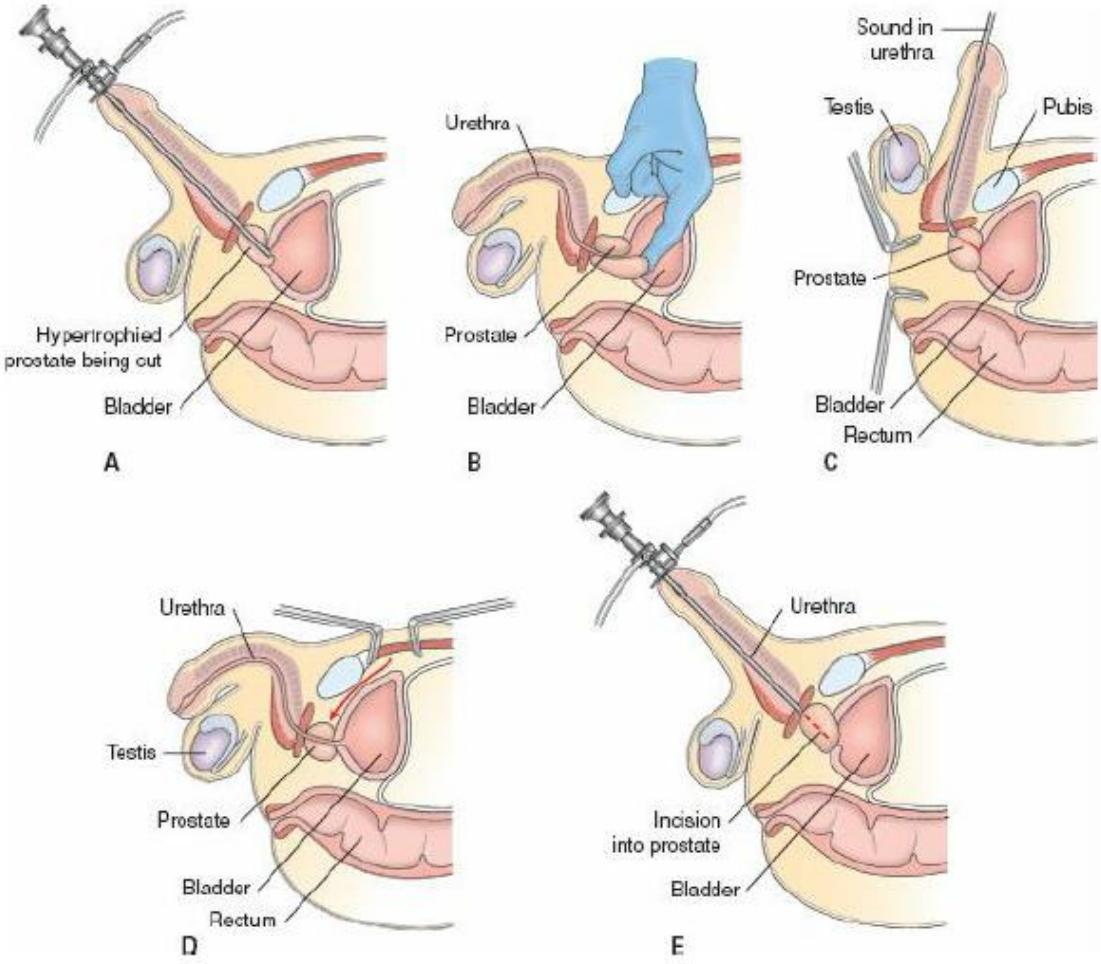
- Cryosurgery of the prostate freezes prostate tissue, killing tumor cells without removing the gland.

Surgical Interventions

- Transurethral Resection of the Prostate
- Suprapubic Prostatectomy
- Perineal Prostatectomy
- Retropubic Prostatectomy
- Transurethral Incision of the Prostate
- Laparoscopic Radical Prostatectomy

Monitoring and Managing Potential Complications

- Hemorrhage
- Infection
- Deep Vein Thrombosis
- Obstructed Catheter



Prostate surgery procedures.

A. Transurethral resection of the prostate. A loop of wire connected with a cutting current is rotated in the cystoscope to remove shavings of prostate at the bladder orifice.
B. Suprapubic prostatectomy. With an abdominal approach, the prostate is shelled out of its bed.
C. Perineal prostatectomy. Two retractors on the left spread the perineal incision to provide a view of the prostate.
D. Retropubic prostatectomy is performed through a low abdominal incision. Note two abdominal retractors and arrow pointing to the prostate gland.
E. Transurethral incision of the prostate involves one or two incisions into the prostate to reduce pressure on the urethra.